

## MATHEMATICS GRADE 10: ROTATION

*CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.2 (10 Lessons)*

| SUB-STRAND OVERVIEW                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <b>Grade Level</b>                         | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Subject</b>                             | Mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Strand</b>                              | Strand 3.0: Measurements and Geometry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Sub-Strand</b>                          | Sub-Strand 3.2: Rotation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Total Duration</b>                      | 10 lessons × 40 minutes = 400 minutes total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Sub-Strand Content</b>                  | <ul style="list-style-type: none"> <li>– Properties of rotation</li> <li>– Angle and centre of rotation given an object and its image</li> <li>– Order of rotational symmetry</li> <li>– Axis and order of rotational symmetry</li> <li>– Congruence and rotation</li> <li>– Applications of rotation in real-life situations</li> </ul>                                                                                                                                                                                                                                                                                                                                                    |
| <b>Learning Outcomes</b>                   | <p>By the end of the sub-strand, the learner should be able to:</p> <ol style="list-style-type: none"> <li>a) state the properties of rotation,</li> <li>b) determine the angle and centre of rotation given an object and its image,</li> <li>c) determine the order of rotational symmetry of plane figures,</li> <li>d) determine the axis and order of rotational symmetry of solids,</li> <li>e) relate congruence to rotation,</li> <li>f) apply rotation in real-life situations,</li> <li>g) appreciate the use of rotation in day-to-day life.</li> </ol>                                                                                                                          |
| <b>Core Competencies</b>                   | <ul style="list-style-type: none"> <li>– Critical Thinking and Problem Solving: the learner locates the centre and angle of rotation from an object and its image and explains why Earth's rotation produces day and night.</li> <li>– Communication and Collaboration: the learner discusses with peers and justifies predictions about rotation using diagrams and correct mathematical language.</li> <li>– Creativity and Imagination: the learner designs rotationally symmetric patterns inspired by Kenyan kiondo weaving, Maasai beadwork and Kikoy designs.</li> <li>– Digital Literacy: the learner uses simulations and GeoGebra to rotate figures and check results.</li> </ul> |
| <b>Core Values</b>                         | <ul style="list-style-type: none"> <li>– Unity: learners work together in groups to model and compare rotations.</li> <li>– Responsibility: learners take care of geometry instruments (compass, protractor, tracing paper) when constructing images.</li> <li>– Integrity: learners report their measurements and evidence honestly.</li> </ul>                                                                                                                                                                                                                                                                                                                                            |
| <b>Science &amp; Engineering Practices</b> | <ul style="list-style-type: none"> <li>– Asking questions and defining problems: learners generate questions about the day and night video for the Driving Question</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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|                                                   | <p>Board.</p> <ul style="list-style-type: none"> <li>– Developing and using models: learners model Earth's rotation with a globe and torch, and rotate figures on the Cartesian plane.</li> <li>– Analyzing and interpreting data: learners measure angles and distances from centre to compare object and image points.</li> <li>– Using mathematics and computational thinking: learners compute rotation angles (e.g. <math>360^\circ</math> in 24 hours = <math>15^\circ</math> per hour).</li> <li>– Constructing explanations: learners use evidence from rotations to explain the day and night pattern.</li> <li>– Engaging in argument from evidence: learners defend predictions about centre, angle and symmetry order.</li> </ul>                                                                                                                                                                                                                                                                                             |
| <b>Pertinent &amp; Contemporary Issues (PCIs)</b> | <ul style="list-style-type: none"> <li>– Life Skills (self-esteem): the learner participates confidently in discussions on rotation.</li> <li>– Education for Sustainable Development: the learner relates rotation to the use of solar energy and time-keeping in daily life.</li> <li>– Learner Support Programmes (critical thinking): the learner solves practical problems involving rotation.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Career Connections</b>                         | <ul style="list-style-type: none"> <li>– Pilot or navigator: uses rotation, bearings and Earth's rotation to understand time zones and flight paths at JKIA.</li> <li>– Mechanical engineer: designs gears, wheels and turbines using rotational symmetry.</li> <li>– Architect and graphic designer: creates symmetric patterns and logos.</li> <li>– Astronomer or meteorologist: studies Earth's rotation, day and night and weather patterns.</li> <li>– Fashion and craft designer: creates rotational patterns in kitenge, kiondo and beadwork.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Focus for Lessons</b>                          | <p>This unit uses the daily cycle of day and night in Kenya as an anchor for understanding rotation as a transformation with a centre, an angle and a direction. Learners predict, observe, explain, record questions on the Driving Question Board and build a growing model, moving from properties of rotation to centre and angle, rotational symmetry, congruence and real-life applications.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Driving Question / Key Inquiry Questions</b>   | <p>DRIVING QUESTION: What do you think makes it possible for a daily pattern of day and night experience on Earth?</p> <p>KICD KEY INQUIRY QUESTIONS:</p> <ol style="list-style-type: none"> <li>1. How do we use rotation in real-life situations?</li> <li>2. Where do we apply rotational symmetry in day-to-day life?</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Anchoring Phenomenon</b>                       | <p>Learners watch a time-lapse video of Earth from space showing day and night sweeping across the globe. It is dawn over Mombasa on the Indian Ocean coast, midday over Nairobi, sunset over Lake Victoria, and darkness settles over Kisumu. The same pattern repeats every 24 hours, and the Sun appears to rise in the east and set in the west at nearly the same times each day. It is puzzling because Kenya feels stationary, yet the video suggests a steady turning. Learners wonder what keeps the cycle regular, why the Sun's position changes, why Mombasa gets sunrise before Kisumu, and how the turning can be described mathematically with a centre, an angle and a direction.</p>                                                                                                                                                                                                                                                                                                                                     |
| <b>Storyline Thread</b>                           | <p>Lesson 1: Phenomenon – learners watch the day and night video, predict what causes the cycle, record questions on the DQB and make an initial model with a globe and torch.</p> <p>Lesson 2: Properties of rotation – learners rotate figures on tracing paper and the Cartesian plane to find that distances from the centre, shape and size are preserved and the direction is turned.</p> <p>Lesson 3: Angle and centre of rotation given object and image – learners use perpendicular bisectors of point pairs to locate the centre, then measure the angle; link to <math>15^\circ</math> per hour for Earth.</p> <p>Lesson 4: Locating the centre and angle on grids – learners practise with positive and negative angles and clockwise and anticlockwise turns, and construct images of given objects.</p> <p>Lesson 5: Order of rotational symmetry of plane figures – learners explore regular polygons, kiondo patterns and wheel designs to count how many times a figure fits onto itself in <math>360^\circ</math>.</p> |

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|                             | <p>Lesson 6: Axis and order of rotational symmetry of solids – learners explore cuboids, cones and prisms using models, including the Earth's axis in the globe model.</p> <p>Lesson 7: Congruence and rotation – learners show that an object and its image are directly congruent and justify with measurements.</p> <p>Lesson 8: Applications in real-life situations – learners analyse wheels, turbines, clocks and patterns in Kenyan crafts and engineering.</p> <p>Lesson 9: Revisiting the phenomenon – learners revise their models and use rotation language (centre, angle, direction, <math>360^\circ</math> in 24 hours) to explain day and night in Mombasa, Nairobi and Kisumu.</p> <p>Lesson 10: Consolidation and assessment – learners complete a performance task designing a rotationally symmetric pattern, explain it, and review the DQB to evaluate how their understanding has grown.</p> |
| <b>Supporting Phenomena</b> | <ul style="list-style-type: none"> <li>– A bicycle wheel on a boda boda spins about its centre, and each spoke returns to its starting position after a full turn.</li> <li>– A wind pump or turbine in Ngong Hills turns blades through equal angles to look identical several times in one turn.</li> <li>– The hands of a clock in a Nairobi classroom rotate about a fixed centre through predictable angles each hour.</li> <li>– A kiondo base or Maasai beaded necklace pattern looks the same after being turned through certain angles.</li> </ul>                                                                                                                                                                                                                                                                                                                                                         |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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## LESSON 1 (40 minutes (1 lesson)): Phenomenon: Day and Night over Kenya

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>Purpose</b>              | To launch the anchoring phenomenon of the regular day and night cycle over Kenya, so that learners see that turning about a centre through an angle in a direction is the mathematical idea they will study in the whole Rotation sub-strand.                                                                                                                                                                                                                                                                                                                                                            |
| <b>Knowledge</b>            | Learners recognise that day and night repeat every 24 hours, that Mombasa, Nairobi and Kisumu see the Sun at different times, and that a turning can be described by three things: a centre, an angle and a direction. They are introduced to the facts that a full turn is $360^\circ$ , a half turn is $180^\circ$ and a quarter turn is $90^\circ$ . Positive angles mean anticlockwise turns and negative angles mean clockwise turns. They make a first, tentative statement that Earth rotates on its own axis, so that some parts face the Sun (day) while the other side is in darkness (night). |
| <b>Skills</b>               | Developing and using models (a globe and torch initial model); asking questions (writing DQB questions); making and recording predictions; obtaining and communicating information from a video (noting times and places); using simple measurement ideas such as $360^\circ \div 24 \text{ h}$ to describe turning.                                                                                                                                                                                                                                                                                     |
| <b>Attitudes</b>            | Curiosity about everyday patterns in Kenya; willingness to make a prediction and revise it later; respect for peers' ideas; collaboration in groups; responsibility in handling the globe and torch; appreciation of the use of mathematics in explaining natural patterns.                                                                                                                                                                                                                                                                                                                              |
| <b>Key Inquiry Question</b> | 1. How do we use rotation in real-life situations? 2. Where do we apply rotational symmetry in day-to-day life? Driving question: What do you think makes it possible for a daily pattern of day and night experience on Earth?                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Purpose in Storyline</b> | This is the launch lesson and builds on no earlier lesson. It creates the shared puzzle: Kenya feels still, but the video suggests a steady turning. Learners' predictions, DQB questions and the first globe-and-torch model will be revisited throughout the unit. Lesson 2 will take the idea of turning about a centre with a direction and study the properties of rotation. Lesson 9 will return to revise this first model.                                                                                                                                                                       |
| <b>Safety Notes</b>         | Never look directly at the Sun. Use a torch only for the model and do not shine it into anyone's eyes. Keep the globe on a table or a stable stand so that it does not roll off. Keep walkways clear while groups darken the room. Handle the globe with both hands and turn it gently.                                                                                                                                                                                                                                                                                                                  |

### B. LESSON OVERVIEW

Teacher Mwangi opens by asking learners whether Kenya is moving. Wanjiku in Kisumu says 'Kenya feels still'. Learners then watch a time-lapse of Earth from space in which 24 hours are shown in 24 seconds (1 second = 1 hour). They see dawn over Mombasa at about 6:30 am EAT, midday over Nairobi at about 12:30 pm, sunset over Lake Victoria at about 6:40 pm and darkness over Kisumu at about 7:30 pm. The pattern repeats every 24 hours. Learners first predict individually what causes the cycle, then observe the video twice and record evidence, then use the globe and a torch to make a first model and explain it in groups. The class opens the Driving Question Board (DQB) with questions about why the Sun's position changes, why Mombasa gets sunrise before Kisumu, and how the turning can be described with a centre, an angle and a direction. Learners finish by drawing and labelling an initial model that will be revised in later lessons.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                | Learner Experience                                             | Teacher Moves                                                       | Sensemaking Strategy                                             | Formative Assessment Strategy                                       |
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| <b>Predict Phase</b> | Learners sit in groups of four. Teacher Mwangi tells the class | Say: 'Wanjiku says Kenya feels still. Before I show you anything, I | Prediction with justification (Predict-Observe-Explain). Writing | Check the prediction notebooks for: (1) a written prediction with a |

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| <p> <b>VIDEO:</b><br/> <a href="#">Determining rotations</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Determine rotations (basic)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p>                   | <p>about Wanjiku in Kisumu, who watches the Sun set over Lake Victoria and says 'Kenya feels still', and Otieno in Mombasa, who sees dawn first over the Indian Ocean. Each learner silently writes in their notebook: (a) What do you think makes the Sun rise and set every day? (b) Who sees sunrise first, Otieno in Mombasa or Wanjiku in Kisumu, and why? (c) Draw a quick sketch of Earth, the Sun and where you think Kenya is during the day and during the night. Learners then share in pairs and one idea per group is noted on the board, such as 'the Sun moves around Earth', 'Earth turns' or 'clouds block the Sun'.</p>                                                                                                                                                                     | <p>want your honest prediction. There are no wrong answers today.' Give 2 minutes of silent writing and then WAIT TIME of 10 seconds after asking: 'What might make the Sun rise and set every single day?' Cold-call 3 learners from different groups to share their prediction, and ask each one: 'What makes you think that?' Then cold-call 2 more learners to share who sees sunrise first. Record all predictions without judging, under the heading 'Our first ideas' on the board. Move around and note which learners already use words such as turn, spin or rotate.</p>                                                                                                                                                                  | <p>a prediction before seeing the evidence makes learners commit to an idea. This brings out prior knowledge and misconceptions (for example that the Sun travels around a still Earth), and it gives them something concrete to compare with their later observations.</p>                 | <p>reason, not just 'I don't know'; (2) a sketch showing Earth, the Sun and Kenya; (3) use of words about motion such as turn, spin, move or rotate. Tally on a clipboard how many learners think the Sun moves and how many think Earth turns, to use as a baseline for Lesson 9.</p>                                                                                                 |
| <p><b>Observe Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Rotating points</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Determine rotations</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Learners watch the time-lapse video of Earth from space, in which 24 hours of Earth time are shown in 24 seconds. First viewing: they just watch. Second viewing: each group records in a table the place, the time (EAT) and what they see: dawn over Mombasa (about 6:30 am), midday over Nairobi (about 12:30 pm), sunset over Lake Victoria (about 6:40 pm) and darkness over Kisumu (about 7:30 pm). They also note the direction in which the light sweeps across the globe and how many times the day-night pattern repeats. They answer: 'How long does it take for the pattern to repeat?' and 'In the video, how many hours of real time pass in 1 second?' (1 second = 1 hour). Learners also answer: 'Which of the three cities, Mombasa, Nairobi or Kisumu, do we see in daylight first?'</p> | <p>Before the video say: 'Watch with scientists' eyes. Look for patterns, not for the answer.' Play the video once with no talking. Then say: 'Turn to your partner and share one thing you noticed' (30 seconds). Play it a second time and pause at each of the four snapshots to let groups fill their tables. After the pause, WAIT TIME of 10 seconds and ask: 'What is the same each time the video repeats?' Cold-call 4 learners for one observation each. Ask: 'If 24 seconds shows one full day, how many seconds would half a day be?' Point out gently that the Sun appears to rise in the east and set in the west at nearly the same times each day, and that Mombasa is on the east coast. Do NOT yet give the full explanation.</p> | <p>Structured observation with a data table and noticing/wondering. Recording times and places turns a vivid video into evidence that can be reasoned with. The 1 second = 1 hour link helps learners connect the video with real time and prepares them for the later rate of turning.</p> | <p>Look at the tables for: the four places and their times recorded correctly (Mombasa dawn about 6:30 am, Nairobi midday about 12:30 pm, Lake Victoria sunset about 6:40 pm, Kisumu dark about 7:30 pm); a statement that the pattern repeats every 24 hours; and a correct reading of 1 second as 1 hour. Note which learners notice that Mombasa is in the light before Kisumu.</p> |

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| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Examples: Determine Angles of Rotation</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Determine rotations (basic)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Groups use a classroom globe tilted about <math>23.5^\circ</math> and a torch as the Sun. One learner holds the torch steady, one finds Kenya (Mombasa, Nairobi, Kisumu) on the globe, one turns the globe gently and one records. They first turn the globe slowly and watch what happens to the lit part at Kenya. They then try turning it in the opposite direction and compare which direction (anticlockwise as seen from above the North Pole) matches the video, where Mombasa meets the light before Kisumu. Each group makes a first explanation: 'We think day and night happen because...' and draws their initial model. They use the words centre, angle and direction in at least one sentence, such as 'The globe turns about its axis (centre), in an anticlockwise direction, through a full turn of <math>360^\circ</math>.'</p> | <p>Say: 'Now you are Earth scientists. Use the torch as the Sun. Make Mombasa, then Nairobi, then Kisumu go from light into dark.' Circulate and ask: 'Which part of the globe is lit? What is the globe turning around?' Prompt with: 'If the torch stays still, what must be moving for Mombasa to get dawn first?' After each question WAIT 10 to 15 seconds before taking answers. Cold-call 3 groups to demonstrate at the front. Introduce, briefly and only as a first idea, that a turning is described by a centre, an angle and a direction, with a full turn <math>360^\circ</math>, a half turn <math>180^\circ</math> and a quarter turn <math>90^\circ</math>; positive angles are anticlockwise and negative angles are clockwise. Say: 'We will test this carefully in the next lessons.' Accept tentative explanations and do not announce the <math>15^\circ</math> per hour rate yet, but if learners ask, say: 'Good question: put it on the board.'</p> | <p>Modelling with physical objects and Think-Pair-Share. Manipulating a globe and torch makes an invisible idea visible: the light stays fixed while the globe turns, so places go from day to night. Talking in groups lets learners test the model against the video evidence and spot where it matches (Mombasa gets light first) or fails.</p> | <p>Observe whether groups: (1) keep the torch still and turn the globe, not the torch; (2) turn the globe anticlockwise when viewed from above the North Pole; (3) show Mombasa reaching the light before Kisumu; (4) use the words centre, angle or direction correctly. Listen for the idea that one half of Earth is always lit and the other half is dark. Record misconceptions such as 'the Sun goes around Earth' for later lessons.</p> |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Determining rotations</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Determine rotations</a><br/> <i>Related topic: not an exact match for this lesson.</i></p>                                                                                                                                                       | <p>The Driving Question is posted on a wall chart: 'What do you think makes it possible for a daily pattern of day and night experience on Earth?' Each learner writes at least one question on a sticky note, such as 'Why does Mombasa get sunrise before Kisumu?', 'Why does the Sun's position change?', 'What keeps the cycle regular every 24 hours?', 'How can we describe the turning using a centre, an angle and a direction?', or 'Where else do we see rotation, such as wheels and wind turbines?' Groups stick the notes on the DQB, and the class sorts them into categories, for example 'Why the cycle repeats', 'Why places differ in time' and 'Describing the turning with</p>                                                                                                                                                     | <p>Say: 'What are you still wondering? A good question starts with how or why.' Give 3 minutes for writing and then WAIT TIME of 10 seconds. Cold-call 5 learners to read out and place their question and ask each: 'Which of our videos or globe observations made you ask that?' Group the notes with the class and add the two Key Inquiry Questions: 'How do we use rotation in real-life situations?' and 'Where do we apply rotational symmetry in day-to-day life?' Tell learners: 'We will return to this board in every lesson to move questions we can answer.' Keep the board in a visible place in the classroom.</p>                                                                                                                                                                                                                                                                                                                                           | <p>Driving Question Board with question sorting. Learners' own questions give ownership of the learning and give a purpose for each later lesson. Categorising questions shows learners how their wonderings link together and which mathematics (angles, direction, rotation) they will need.</p>                                                 | <p>Check the sticky notes: every learner wrote at least one question; questions are on topic and phrased as how or why; at least some link to the phenomenon (Mombasa before Kisumu, regularity, the Sun's position) or to turning with a centre, angle and direction. Note learners who need help to turn a statement into a question.</p>                                                                                                     |

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| Source: Khan Academy (English - US curriculum)<br><a href="#">Search ARES for similar readings</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | mathematics'. Learners link their questions to the two Key Inquiry Questions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Model Building Phase</b><br> <b>VIDEO:</b><br><a href="#">Rotating points</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br><a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Determine rotations (basic)</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br><a href="#">Search ARES for similar readings</a> | Each learner draws an initial model in their notebook or on a model sheet. The model has: Earth (as a circle or globe) with a dot for the axis (the centre of turning), the Sun on one side with light arrows, Kenya marked with Mombasa, Nairobi and Kisumu, a curved arrow for the direction of turning (anticlockwise as viewed from above the North Pole), and labels for day and night. Under the drawing they write a caption: 'I think day and night happen because...' and list one thing they are still unsure about. Groups compare models and tick what is the same. Models are kept in a folder, to be revised in Lesson 9. | Say: 'Draw what is happening, then explain it in words. Include a centre, a direction, and where the light comes from.' Circulate and ask at least 6 learners: 'Where is the centre of the turning in your model?' and 'Which way is it turning?' WAIT 10 seconds for answers. Cold-call 2 learners to show their model under the document camera or hold it up. Close by saying: 'Today you have a first model. Mathematics will help you sharpen it: we will study how turning works, how to measure it, and later how many degrees Earth turns each hour.' Collect the models. | Initial model construction (develop a model, then revise it later). A first drawn and written model records current thinking and exposes gaps. Because it will be revised in Lesson 9, learners experience science and mathematics as an ongoing process of improving explanations with evidence. | Check each model for: Earth with an axis or centre marked; the Sun and light direction; a turning arrow with a direction; Mombasa, Nairobi and Kisumu placed with Mombasa on the east side; a caption with at least one reason; and one stated uncertainty. Note models showing the Sun moving around a still Earth as a misconception to address in later lessons. |

## D. TEACHER REFLECTION

Revisit the Driving Question and ask: What evidence did learners gather today (the 24-hour pattern, the four snapshot times, Mombasa in the light before Kisumu, the globe and torch behaviour)? Which learners moved from 'the Sun moves' to 'Earth turns', and which still hold the idea that Kenya is still? Did every learner contribute a DQB question? Which questions can be answered using the tools of rotation (centre, angle, direction) in the next lessons? Note which learners struggled to handle the globe and torch safely or to use the terms clockwise and anticlockwise, and plan support. Consider whether the video was too fast for some groups and whether to replay it in Lesson 2.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                            |                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What did I observe?</b> | We watched a time-lapse of Earth where 24 hours were shown in 24 seconds (1 second = 1 hour). We saw dawn over Mombasa at about 6:30 am, midday over Nairobi at about 12:30 pm, sunset over Lake Victoria at about 6:40 pm and darkness over Kisumu at about 7:30 pm, and the pattern repeated every 24 hours. With a globe and torch, we saw that turning the globe makes places move from light into dark. |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                              |                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What did I learn?</b>                     | A turning can be described using a centre, an angle and a direction. A full turn is $360^\circ$ , a half turn is $180^\circ$ and a quarter turn is $90^\circ$ . Positive angles are anticlockwise and negative angles are clockwise. Kenya feels still, but the video suggests a steady turning, and Mombasa meets the light before Kisumu.  |
| <b>How does this explain the phenomenon?</b> | Our first explanation is that Earth turns about its axis so that some parts face the Sun and have day while the other side is dark and has night. We still need to find out why Mombasa gets sunrise before Kisumu and how to describe the turning with a centre, an angle and a direction. Our questions are on the Driving Question Board. |

## LESSON 2 (40 minutes (1 lesson)): Properties of Rotation: What Stays the Same and What Turns?

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | Learners investigate what changes and what stays the same when a figure is rotated about a centre, so they can describe the turning of Earth (and Kenya with it) using a centre, an angle and a direction.                                                                                                                                                                                                                                                                                                                                                            |
| <b>Knowledge</b>            | Rotation turns every point of an object about a fixed centre through the same angle in the same direction. Distances from the centre, lengths, angles, shape and size are preserved, so the image is congruent to the object. Positive angles are anticlockwise and negative angles are clockwise. Worked example: with centre $O(0,0)$ , $A(2,1)$ goes to $A'(-1,2)$ , $B(4,1)$ to $B'(-1,4)$ and $C(4,3)$ to $C'(-3,4)$ under a $90^\circ$ anticlockwise turn, where $(x,y)$ goes to $(-y,x)$ . $OA = OA' = \sqrt{5}$ , $OB = OB' = \sqrt{17}$ and $OC = OC' = 5$ . |
| <b>Skills</b>               | Developing and using models (rotating figures with tracing paper and on the Cartesian plane); Planning and carrying out investigations (measuring distances and sides); Analysing and interpreting data (comparing object and image measurements); Constructing explanations (using evidence to state the properties of rotation).                                                                                                                                                                                                                                    |
| <b>Attitudes</b>            | Curiosity about how turning can be described mathematically; collaboration and unity when working in groups; accuracy and responsibility when handling geometry instruments; appreciation of rotation in day-to-day life.                                                                                                                                                                                                                                                                                                                                             |
| <b>Key Inquiry Question</b> | How do we use rotation in real-life situations?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Purpose in Storyline</b> | In Lesson 1 learners met the idea of turning about a centre (Earth's axis) with a direction. Lesson 2 gives that idea mathematical precision by finding what rotation preserves (distance from the centre, shape, size) and what it changes (position and direction of facing). This prepares for finding the centre and angle (Lesson 3) and for explaining why Kenya, though it feels still, keeps its shape and distances while turning with Earth.                                                                                                                |
| <b>Safety Notes</b>         | Handle compasses and pencils with sharp points carefully and keep them pointing downward. Keep pins or compass points away from faces. Do not use a pin through tracing paper unless the teacher demonstrates it, and return all instruments to the box at the end. Walk, do not run, when moving to the DQB wall chart.                                                                                                                                                                                                                                              |

### B. LESSON OVERVIEW

Anchoring phenomenon: the time-lapse shows day and night sweeping over Mombasa, Nairobi and Kisumu every 24 hours, so Kenya, which feels still, is actually turning with Earth. Teacher Mwangi asks: 'If Kenya turns with Earth, do Nairobi and Kisumu get bigger, smaller or change shape?' Learners predict, then rotate a triangle on tracing paper about a centre and on the Cartesian plane ( $O(0,0)$ ,  $A(2,1)$ ,  $B(4,1)$ ,  $C(4,3)$  rotated  $90^\circ$  anticlockwise). They measure distances from the centre and the sides of the triangle, and discover the properties of rotation. They explain these in terms of centre, angle and direction, update the Driving Question Board (DQB) and revise the class model of Earth's turning.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                | Learner Experience                                        | Teacher Moves                                                   | Sensemaking Strategy                                           | Formative Assessment Strategy                                      |
|----------------------|-----------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Predict Phase</b> | Learners recall the time-lapse video from Lesson 1, where | Display a picture of the Earth with Kenya marked and say: 'Last | Predict-Observe-Explain (POE) with a sketch-and-share routine: | Look for: (1) sketches showing the triangle in a new position; (2) |

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| <p> <b>VIDEO:</b><br/> <a href="#">Determining rotations</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Rigid transformations: preserved properties</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Mombasa has dawn, Nairobi midday, Lake Victoria sunset and Kisumu darkness within one 24-hour turn. In pairs they look at a triangle drawn on a sheet and make a prediction in their notebooks: 'If this triangle is turned 90° anticlockwise about a point O, which things will change and which will stay the same: the distance of each corner from O, the side lengths, the size, the shape, the position?' They draw a quick sketch of where they think the triangle will land and write one sentence connecting this to Kenya turning with Earth about its axis.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>lesson we said Kenya is turning about a centre, Earth's axis. Today we ask what stays the same when something turns.' Ask: 'If Nairobi and Kisumu both turn with Earth, does the distance between them change?' Give WAIT TIME of 10-15 seconds before anyone speaks. Then say: 'Turn to your partner and make a prediction with a sketch.' Circulate and cold-call 4 pairs, asking each: 'What do you think stays the same, and why?' Record all predictions on the board without judging them. Remind learners that Wanjiku in Kisumu says 'Kenya feels still' and ask: 'Does it feel still because nothing changes in shape or distance, even though it turns?'</p>                                                                                                                                                                                                                              | <p>committing to a prediction makes learners' existing ideas visible, creates a need to test them, and lets them later compare evidence against their own thinking.</p>                                                                                           | <p>learners who list something that changes (position, direction of facing) and something that stays the same; (3) learners who use the words 'centre', 'angle' or 'direction'. Note common ideas such as 'the triangle gets bigger when it turns' or 'the corners move closer to O' for later testing. Tally how many pairs predict that size stays the same.</p>                                                   |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Rotating points</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Rotate shapes</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p>      | <p>Activity A (tracing paper): In groups of 4, learners draw triangle ABC with A(2,1), B(4,1), C(4,3) on grid paper with the centre O(0,0). They trace the triangle and the axes onto tracing paper, hold O with a pencil point, and turn the tracing paper 90° anticlockwise (a quarter turn) to mark A', B' and C'. They read the image coordinates: A'(-1,2), B'(-1,4) and C'(-3,4). Activity B (measure and check): They measure OA and OA', OB and OB', OC and OC' with a ruler (or compute with Pythagoras) and record a table: OA = OA' = <math>\sqrt{5}</math>, OB = OB' = <math>\sqrt{17}</math>, OC = OC' = 5. They also measure the sides: AB = 2, BC = 2, AC = <math>\sqrt{8}</math> and A'B' = 2, B'C' = 2, A'C' = <math>\sqrt{8}</math>, and check that the angles match. Activity C (direction): They check the rule (x,y) goes to (-y,x) for each point, and then turn the tracing paper clockwise (-90°) to see where A goes: A''(1,-2). They compare the two directions and record what is</p> | <p>Hand out grid paper, tracing paper and rulers and say: 'Hold the centre O steady with your pencil and turn only the paper, as Earth turns about its axis.' Model the first point A on the board. Ask: 'Which way is anticlockwise? Show me with your hand.' and wait for all hands. Give WAIT TIME of 10-15 seconds after asking: 'Compare OA and OA'. What do you notice?' Cold-call 3 groups to give their measurements and write them in a class table. Ask: 'Is the triangle bigger, smaller or the same? How do you know?' Prompt groups that are stuck with: 'Count the squares from O to A, then from O to A'.' Ask a group to demonstrate the negative turn: 'Turn the paper 90° clockwise. Where does A land? What sign do we give this angle?' Circulate and check that learners hold the centre fixed. Remind: 'Positive angle means anticlockwise, negative angle means clockwise.'</p> | <p>Structured data table with pattern-finding (Analysing and interpreting data): recording object and image measurements side by side lets learners see that every pair matches, so they can generalise from repeated cases instead of from a single example.</p> | <p>Observable indicators: learners keep the centre O fixed while turning the tracing paper; they write A'(-1,2), B'(-1,4), C'(-3,4) correctly; their table shows OA = OA', OB = OB', OC = OC'; they state that AB = A'B', BC = B'C' and AC = A'C'; they correctly give A''(1,-2) for -90°. Use a checklist to note which groups mix up clockwise and anticlockwise, and re-model the direction for those groups.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | the same and what is different.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Rotating shapes</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Rigid transformations: preserved properties</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>Groups use their table to write a claim, evidence and reasoning statement: 'Rotation keeps _____ the same and changes _____.' They list the properties: (1) each point stays the same distance from the centre (<math>OA = OA'</math>, <math>OB = OB'</math>, <math>OC = OC'</math>); (2) lengths and angles are preserved, so the shape and size are the same, and the image is congruent to the object; (3) every point turns through the same angle in the same direction about the centre; (4) the position and the direction the figure faces change. They explain why Kenya does not stretch or shrink as it turns: Nairobi, Mombasa and Kisumu keep their distances from Earth's axis and from one another. Each learner writes a sentence describing Earth's turning using centre (Earth's axis), angle (<math>15^\circ</math> per hour) and direction (anticlockwise as seen from above the North Pole).</p> | <p>Say: 'Use your numbers as evidence. Complete the sentence: rotation keeps ... the same.' Give WAIT TIME of 10-15 seconds, then cold-call 5 learners for one property each and write them on the board under the headings 'Same' and 'Changes'. Push reasoning: 'Why must OA equal OA'? What is fixed while the paper turns?' (The centre is fixed and the paper does not stretch.) Ask: 'Otieno in Mombasa and Wanjiku in Kisumu both turn with Earth. Does Otieno get closer to Wanjiku? Why or why not?' Clarify the language: 'Use the word rotation for Earth's spin, not revolution.' Clear up the misconception that a rotated image is a different size or shape, using the measured sides 2, 2 and <math>\sqrt{8}</math>. Ask a final question: 'What three things do we need to describe any rotation?' and gather the response: centre, angle, direction.</p> | <p>Claim-Evidence-Reasoning (CER) with a class consensus discussion: learners link their measured data (evidence) to the general properties (claim) using the idea of a fixed centre and a rigid turn (reasoning), which builds a causal explanation they can apply to Earth.</p> | <p>Use a 3-question exit check on mini whiteboards or notebooks: (1) Name two things that stay the same in a rotation. (2) If <math>OB = \sqrt{17}</math>, what is <math>OB'</math>? (3) What does <math>-90^\circ</math> mean? Look for: 'distance from centre and size/shape', <math>OB' = \sqrt{17}</math>, and '90° clockwise'. Aim for at least 80% correct on questions 1 and 3; plan a reteach with the table for those who are not yet secure. Check that explanations of Earth's turning mention centre, angle and direction.</p> |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Determining rotations</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Rotate shapes</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar</p>  | <p>Learners revisit the DQB wall chart with the driving question 'What makes it possible for a daily pattern of day and night experience on Earth?' They add at least one sticky note each with either a new answer from today's evidence (for example 'Earth turns without changing its shape or size, so the distances between Nairobi and Kisumu stay the same') or a new question (for example 'How do we find the centre if we only see the object and the image?' or 'How much does Earth turn in one hour?'). They place notes next to related questions from Lesson 1 and mark any question that is now</p>                                                                                                                                                                                                                                                                                                      | <p>Say: 'Walk to the DQB with one sticky note. Write what today's rotation evidence helps us answer, or a question it raises.' Give WAIT TIME of 10-15 seconds before groups write. Cold-call 4 learners to read a note aloud and ask: 'Which question on the board does this connect to?' Group notes into categories: centre, angle, direction and what stays the same. Point to the question about Mombasa sunrise before Kisumu and say: 'We will come back to this. What do we still need to know?' Highlight questions about finding the centre and the angle, which lead to Lesson 3. Keep the DQB visible</p>                                                                                                                                                                                                                                                      | <p>Driving Question Board (question tracking): connecting new evidence to the class's own questions shows learners how each lesson moves them closer to answering the driving question and gives purpose to the next investigation.</p>                                           | <p>Check that each learner has contributed a note; the notes should link today's evidence (distances, shape and size preserved, direction of turn) to the phenomenon. Observe whether new questions are mathematical and testable, for example about finding the centre or the angle. Note learners who only restate Lesson 1 questions and prompt them with: 'What did you measure today that could help?'</p>                                                                                                                            |

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| readings                                                                                                                                                                                                                                                                                                                                                                                                                      | partly answered.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | and tell learners that it is a record of their growing understanding.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Model Building Phase</b><br><b>VIDEO:</b><br><a href="#">Rotating points</a><br>Source: Khan Academy (English - US curriculum)<br><a href="#">Search ARES</a> for similar videos<br><br><b>EXERCISE:</b><br><a href="#">Determine rotations (basic)</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br><a href="#">Search ARES</a> for similar readings | Learners return to the class model from Lesson 1 (globe tilted at about $23.5^\circ$ with a torch as the Sun). Using the globe, they mark Nairobi, Mombasa and Kisumu with small stickers and turn the globe anticlockwise (viewed from above the North Pole) to see that the three stickers stay at the same distance from the axis and from each other as the globe turns, which matches the properties found today. In their notebooks they revise their model diagram by adding: the centre (Earth's axis), an arrow showing the direction (anticlockwise), a note that all places turn through the same angle together, and a statement that distances and shapes stay unchanged. They also record a short summary: 'Kenya feels still because it turns with Earth without changing its shape or size.' | Say: 'Take your Lesson 1 model. What does today's evidence let us add or change?' Ask a group to turn the globe slowly anticlockwise while the class watches the three stickers, and ask: 'Do Mombasa and Kisumu move closer to each other? What stays the same?' Give WAIT TIME of 10-15 seconds before taking answers. Cold-call 3 learners to share a revision they made and why. Ask them to label the three parts of the description: 'Where is the centre? Which way is the direction? We will find the angle per hour in later lessons.' Do not give the $15^\circ$ per hour rate as new content today beyond reminding that it is part of the established model. Encourage learners to mark what is still unknown and to add it to the DQB. | Iterative model revision (Developing and using models): updating the same model with new evidence shows how understanding grows and links the abstract properties of rotation to the concrete globe-and-torch model of day and night. | Look for revised diagrams that include a labelled centre (axis), a direction arrow (anticlockwise), and a note that distances, shape and size stay the same. Success indicators: learners can explain why Mombasa and Kisumu stay the same distance apart while turning; they use the words 'centre', 'direction' and 'angle' correctly. Collect notebooks or sample 6 diagrams to identify who needs support with the centre and direction. |

## D. TEACHER REFLECTION

Reflect on the lesson: Which properties of rotation did learners find through their own measurements, and which did I have to tell them? How many learners confused clockwise and anticlockwise or the sign of the angle, and how will I address this in Lesson 3? Did learners connect equal distances  $OA = OA'$  to the idea that the centre is fixed, which will help them find the centre by perpendicular bisectors next lesson? Did the globe model help them see that Kenya turns without changing shape or distance? Which DQB questions are ready to be revisited?

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                     |                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What did I observe? | We rotated triangle ABC with A(2,1), B(4,1), C(4,3) through $90^\circ$ anticlockwise about O(0,0) on tracing paper and on the Cartesian plane, and we obtained A'(-1,2), B'(-1,4), C'(-3,4). We measured $OA = OA' = \sqrt{5}$ , $OB = OB' = \sqrt{17}$ , $OC = OC' = 5$ , and sides $AB = A'B' = 2$ , $BC = B'C' = 2$ , $AC = A'C' = \sqrt{8}$ . A turn of $-90^\circ$ (clockwise) sent A to A''(1,-2). |
| What did I learn?   | A rotation turns every point about a fixed centre through the same angle in the same direction. Distances from the centre, lengths, angles, shape and size stay the same, so the image is congruent to the object. Only the position and the direction the figure faces change. Positive angles are anticlockwise and negative angles are clockwise.                                                     |

**How does this explain the phenomenon?**

Kenya feels still but turns with Earth about its axis. Because rotation preserves distances, shape and size, Nairobi, Mombasa and Kisumu keep their distances from the axis and from one another as they turn anticlockwise together, which helps explain why the day-night pattern repeats regularly. The turning is described with a centre (Earth's axis), an angle and a direction.

## LESSON 3 (80 minutes (double lesson)): Finding the Centre and Angle of Rotation from an Object and Its Image

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | Learners locate the centre of a rotation by constructing perpendicular bisectors of object-image point pairs, then measure the angle and state the direction. They use the same three descriptors (centre, angle, direction) to describe how Earth turns at $15^\circ$ per hour, which explains the day-night pattern over Mombasa, Nairobi and Kisumu.                                                                                                                                                                                                      |
| <b>Knowledge</b>            | The centre of rotation is equally far from a point and its image (Lesson 2: $OA = OA'$ ), so it lies on the perpendicular bisector of each object-image pair. The centre is where at least two different bisectors meet. The angle of rotation is measured from the centre to the object point and then from the centre to the image point. Positive means anticlockwise and negative means clockwise. At $15^\circ$ per hour, a $90^\circ$ turn takes 6 hours.                                                                                              |
| <b>Skills</b>               | Constructing perpendicular bisectors with a ruler and compass. Plotting points on a grid. Measuring angles with a protractor. Using tracing paper to check a rotation. Using Analysing and Interpreting Data and Using Mathematics (NGSS practices) to link an angle to an Earth-time calculation. Constructing Explanations from evidence.                                                                                                                                                                                                                  |
| <b>Attitudes</b>            | Precision and patience in constructions. Collaboration and respect for peers' reasoning. Curiosity about how mathematics describes the turning Earth beneath Kenya.                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Key Inquiry Question</b> | How do we use rotation in real-life situations? (Linked to: if we only see a before and after, how can we work out the centre and angle of the turn, as scientists do for Earth?)                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Purpose in Storyline</b> | Lesson 1 launched the phenomenon and the idea of turning about a centre (Earth's axis) with a direction. Lesson 2 showed that rotation keeps distances from the centre. Lesson 3 uses that idea in reverse: given an object and image, learners find the centre and the angle. The angle-per-hour link ( $90^\circ = 6$ hours at $15^\circ$ per hour) gives them the mathematical language for Lesson 9, where they explain why Mombasa sees sunrise about 20 minutes before Kisumu. Lesson 4 will then construct images using positive and negative angles. |
| <b>Safety Notes</b>         | Compass points and protractor edges are sharp: carry them point-down and keep them away from faces. Pass the globe and torch with two hands. Do not shine the torch into anyone's eyes. Keep walkways clear when learners move between groups.                                                                                                                                                                                                                                                                                                               |

### B. LESSON OVERVIEW

Teacher Mwangi replays the 24-second time-lapse and asks the class to find the centre of Earth's turning from two snapshots only. Learners predict how a centre could be found when only an object and its image are given. In the Observe phase they work in groups on grid paper with  $P(2,0) \rightarrow P'(0,2)$ ,  $Q(4,0) \rightarrow Q'(0,4)$  and  $R(3,2) \rightarrow R'(-2,3)$ . They construct perpendicular bisectors, discover that the bisectors of  $PP'$  and  $QQ'$  fall on the same line ( $y = x$ ), and so need the bisector of  $RR'$  ( $y = 5x$ ) to pin down the centre. The bisectors of  $PP'$  and  $RR'$  meet at  $O(0,0)$ . They then measure angle  $POP'$  as  $90^\circ$  anticlockwise, which is  $+90^\circ$ . In the Explain phase they convert  $90^\circ$  into 6 hours using  $15^\circ$  per hour and connect it to the video: 6:30 am dawn over Mombasa to 12:30 pm midday over Nairobi is 6 hours, so Earth has turned  $90^\circ$ . They add new questions to the DQB and add the centre-angle-direction method to their globe-and-torch model.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| <p><b>Predict Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Determining rotations</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Determine rotations (basic)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p>                         | <p>Learners watch the 24-second time-lapse again (1 second = 1 hour = <math>15^\circ</math> of turning): dawn over Mombasa about 6:30 am, midday over Nairobi about 12:30 pm, sunset over Lake Victoria about 6:40 pm, darkness over Kisumu about 7:30 pm. Teacher Mwangi shows two paused frames, an early one and a later one, with the Indian Ocean coast in different places. Learners individually write a prediction in their books: 'If I only had a before picture and an after picture, how could I find the point the whole thing turned around, and how could I find how far it turned?' They then pair up and sketch their method. Wanjiku's remark 'Kenya feels still' is put on the board: 'If Kenya is turning, where is the centre?'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Say: 'Nothing on this screen has a label for the centre. Using only the before and after positions, how would you find it?' Give 2 minutes of silent writing. WAIT TIME of 10-15 seconds after asking, with no hints. Then cold-call 4 learners, asking each to read their prediction without correcting. Probe with: 'What did Lesson 2 tell us about the distance from the centre to a point and to its image?' Record 3 or 4 predictions on the board, for example 'the centre is in the middle', 'join the points and find the middle', 'measure with a protractor'. Do not say which is correct.</p>                                                                                                                                                                                                                                         | <p>Predict-Observe-Explain (POE) with Think-Pair-Share. Writing a prediction before instruction exposes learners' existing ideas, including the common misconception that the centre is the midpoint of the line joining a point to its image. Learners will later return to these predictions with evidence, which builds ownership of the conclusion.</p>                                                                                                                                               | <p>Look for learners who mention equal distances from the centre (Lesson 2 idea) or a 'middle line' between a point and its image. Tally on a clipboard how many predictions rely on the midpoint only and how many mention the centre being equally far from both. Note learners who cannot suggest any method and give them extra support in the Observe phase.</p>                                                                                                                                                                                                                      |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Geometric constructions: perpendicular bisector</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Determine rotations</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>In groups of 4 with grid paper, rulers, compasses, protractors and tracing paper, learners plot <math>P(2,0)</math>, <math>P'(0,2)</math>, <math>Q(4,0)</math>, <math>Q'(0,4)</math>, <math>R(3,2)</math> and <math>R'(-2,3)</math>. They draw the segments <math>PP'</math>, <math>QQ'</math> and <math>RR'</math>. For each, they construct the perpendicular bisector: open the compass past half the segment length, draw arcs from both ends, and join the two crossing points. Step 1: they find that the bisector of <math>PP'</math> is the line <math>y = x</math>. Step 2: they find the bisector of <math>QQ'</math> also lies on <math>y = x</math>, so the two bisectors lie on the same line and give no single meeting point. Step 3: they draw the bisector of <math>RR'</math> (the line <math>y = 5x</math>) and mark where it crosses <math>y = x</math>, which is <math>O(0,0)</math>. Step 4: they check by measuring <math>OP = OP' = 2</math> and <math>OQ = OQ' = 4</math>. Step 5: they place a protractor at <math>O</math>, measure from <math>OP</math> to <math>OP'</math> and find <math>90^\circ</math> anticlockwise, so the angle is <math>+90^\circ</math>, and they</p> | <p>Circulate and ask: 'Why do you need a third pair?' Wait 10-15 seconds for each group to answer. Say: 'Where do two lines have to meet to give one centre?' When groups say 'the bisectors of <math>PP'</math> and <math>QQ'</math> are the same line', respond: 'Good noticing, so what does that tell us about how many pairs we need, and what must be true about the lines?' Cold-call 3 groups to show where their bisectors cross. Pause the class at <math>O</math> and say: 'Put your protractor at <math>O</math>. From <math>OP</math> to <math>OP'</math>, which way did we turn and how far? Is that positive or negative?' Use a tracing paper demonstration: pin at <math>O</math>, turn the paper anticlockwise until <math>P</math> lands on <math>P'</math>. Then ask: 'Which of our predictions held up, and which did not?'</p> | <p>Evidence-gathering with structured data collection (Using Mathematics and Analysing Data practice). Learners build a table: pair, midpoint, equation of bisector, distance to centre. Constructing and then checking with tracing paper turns a rule ('the centre is on the perpendicular bisector') into something they have seen work. Discovering that two pairs can share a bisector, and that a third pair is needed, builds understanding of why at least two different bisectors must meet.</p> | <p>Observable indicators: (1) bisectors drawn at right angles through the midpoints, accurate to within about 2 mm; (2) the learner explains why the bisectors of <math>PP'</math> and <math>QQ'</math> do not give a single point; (3) the centre is correctly marked at <math>O(0,0)</math>; (4) the angle is read from the correct scale and stated as <math>+90^\circ</math> or <math>90^\circ</math> anticlockwise; (5) tracing paper confirms that <math>P</math> maps to <math>P'</math>. Record on a checklist which groups needed help with compass arcs or protractor scale.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | check with tracing paper pinned at O. Finally, they compute that $90^\circ \div 15^\circ$ per hour = 6 hours.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Determining rotations</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Determine rotations (basic)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Learners write a three-part description of the rotation: centre O(0,0), angle <math>90^\circ</math>, direction anticlockwise (<math>+90^\circ</math>). They then connect it to the phenomenon. Earth turns <math>15^\circ</math> per hour, so a turn of <math>90^\circ</math> takes <math>90 \div 15 = 6</math> hours. In the video, 6:30 am dawn over Mombasa to 12:30 pm midday over Nairobi is 6 hours, so between those two snapshots Earth turned through <math>90^\circ</math>. They fill in a sentence frame: 'The centre of Earth's turning is ____, the angle between the snapshots is ____ <math>^\circ</math>, and the direction is ____ as seen from above the North Pole.' They also answer: 'If the angle were <math>45^\circ</math>, how many hours would pass?' (3 hours). Finally each group writes a claim, evidence and reasoning (CER) paragraph on why the centre can be found without being seen.</p> | <p>Ask: 'In the globe model, what plays the role of the centre O?' Wait 10-15 seconds, then take answers from 3 cold-called learners (Earth's axis). Ask: 'Why is a <math>90^\circ</math> turn about 6 hours and not 24?' Wait 10-15 seconds. Cold-call 3 learners to show the division. Use the globe and torch: turn the globe anticlockwise (viewed from above the North Pole) through <math>90^\circ</math> and ask where Mombasa and Kisumu are relative to the light. Remind learners: 'The Sun only appears to move east to west; it is Kenya that turns.' Challenge: 'Does the centre depend on the pair you choose?' Pairs with the same bisector support the same centre, so only different bisectors can fix it.</p> | <p>Claim-Evidence-Reasoning (CER) paired with Mathematical Modelling. Learners link a geometric result (<math>+90^\circ</math> about O) to a physical rate (<math>15^\circ</math> per hour) and a real observation (6:30 am to 12:30 pm). This transfer from a grid problem to Earth's turning turns the abstract construction into a tool for describing a natural phenomenon.</p> | <p>Check CER paragraphs for: a clear claim (the centre lies where perpendicular bisectors meet); evidence from the construction (<math>OP = OP' = 2</math>, bisectors meeting at O); reasoning (equal distances put the centre on each bisector). Check the conversion <math>90^\circ \div 15 = 6</math> h. Note learners who confuse the 6-hour gap with a 24-hour turn and revisit with a <math>360^\circ \div 24</math> h calculation.</p> |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Geometric constructions: perpendicular bisector</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Determine rotations</a><br/> <i>Related topic: not an exact match</i></p>                                                                                                                                                              | <p>Learners return to the DQB wall chart with sticky notes. Each pair adds at least one new question and moves any question they can now answer to an 'Answered by evidence' column. Examples: 'Is Earth's centre of turning a point or a line?' 'If Mombasa is <math>2.9^\circ</math> east of Nairobi, how many minutes earlier does it get the Sun?' 'How would we find a centre if the image were turned clockwise?' They add evidence cards: 'Perpendicular bisectors of PP' and RR' meet at O(0,0). Turn of <math>+90^\circ</math> takes 6 hours at <math>15^\circ</math> per hour.' Learners cluster questions by theme (centre, angle, direction, timing in Kenya).</p>                                                                                                                                                                                                                                                 | <p>Say: 'Which question on our board can we now answer, and what is the evidence?' WAIT TIME of 10-15 seconds. Cold-call 4 learners to read and place a note. Say: 'Which questions need Lesson 4's grids and which need our globe model?' Point to Otieno's question about dawn at Mombasa and say: 'We now have the <math>15^\circ</math> per hour tool to calculate this later.' Do not answer all questions: leave the Mombasa-before-Kisumu timing open for Lesson 9.</p>                                                                                                                                                                                                                                                  | <p>Driving Question Board (DQB) with question clustering. Seeing questions move from 'open' to 'answered by evidence' shows learners that the mathematics is advancing the driving question 'what makes day and night possible?' Clustering questions helps learners decide what evidence is still needed.</p>                                                                      | <p>Count new questions that involve the centre, angle or direction. Check that learners cite specific evidence (the bisector meeting point, 6 hours for <math>90^\circ</math>). Note whether questions show progress, for example from 'Why does the Sun move?' to 'How many degrees does Earth turn between Mombasa dawn and Nairobi midday?'</p>                                                                                            |

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| <p>for this lesson.</p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                    |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b></p> <p><a href="#">Rotating points</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b></p> <p><a href="#">Determine rotations (basic)</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar readings</a></p> | <p>Learners revise the initial globe-and-torch model from Lesson 1. On a poster in their groups, they draw Earth from above the North Pole with the axis as the centre. They mark Mombasa, Nairobi and Kisumu, an arrow showing anticlockwise turning, and a label '15° per hour'. They add the 90° turn between the video snapshots at 6:30 am and 12:30 pm as an angle at the centre, labelled '+90° = 6 h'. A short note under the poster lists the method: 'Find centre by bisectors, measure angle from centre to object and to image, state direction.' Using the globe and torch, they turn the globe 90° anticlockwise and describe which Kenyan places move into the light.</p> | <p>Say: 'Add one thing to your model that you could not draw in Lesson 1.' WAIT TIME of 10-15 seconds, then circulate. Ask each group: 'Where is the centre in your model, and what is the angle for 6 hours?' Cold-call 3 groups to present their revised model in 30 seconds each. Prompt: 'What is still missing for us to explain why Mombasa sees sunrise before Kisumu?' (the longitude difference and timing, coming later.) Leave posters on the wall for later lessons.</p> | <p>Cumulative Model Revision with Gallery Walk. Revising the same model each lesson shows how new evidence (a method for centre and angle, and the 15° per hour rate) improves explanatory power. Naming what is still missing keeps the focus on the driving question.</p> | <p>Check that each model shows: the axis as the centre, an anticlockwise arrow, '15° per hour', and the 90° turn labelled as 6 hours. A strong model labels all three descriptors (centre, angle, direction). Record any group that places the centre outside Earth or reverses the direction.</p> |

## D. TEACHER REFLECTION

Which learners confused the midpoint of PP' with the centre, and did the tracing-paper check correct this? Did groups notice that the bisectors of PP' and QQ' lie on the same line, and could they explain why a third pair was needed? How well did learners connect 90° with 6 hours using 15° per hour? Which parts of the construction (compass arcs, protractor reading) need more practice before Lesson 4? What evidence on the DQB suggests learners are moving toward explaining the Mombasa-Kisumu sunrise difference?

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                            |                                                                                                                                                                                                                                                                                                                                                                |
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| <b>What did I observe?</b> | We constructed the perpendicular bisectors of PP', QQ' and RR' on a grid. The bisectors of PP' and RR' met at O(0,0), and the bisector of QQ' lay on the same line as that of PP' ( $y = x$ ). Tracing paper confirmed that turning through 90° anticlockwise about O takes P to P'. In the video, 6:30 am Mombasa dawn to 12:30 pm Nairobi midday is 6 hours. |
| <b>What did I learn?</b>   | The centre of rotation lies where the perpendicular bisectors of object-image pairs meet, and at least two different bisectors are                                                                                                                                                                                                                             |

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|                                              | needed. The angle is measured from the centre to the object point and then to the image point, positive if anticlockwise. Earth turns 15° per hour, so 90° takes 6 hours.                                                                                                                                                     |
| <b>How does this explain the phenomenon?</b> | Earth's rotation can be described by a centre (its axis), an angle (15° per hour) and a direction (anticlockwise as seen from above the North Pole). Kenya is not still: it turns with Earth, and that steady turning brings Mombasa, Nairobi and Kisumu in turn into the Sun's light, creating day and night every 24 hours. |

## LESSON 4 (40 minutes): Rotations on Grids: Positive and Negative Angles, Clockwise and Anticlockwise Turns

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <b>Purpose</b>              | Learners construct images of objects on a grid using positive (anticlockwise) and negative (clockwise) angles, and use the coordinate rules for $90^\circ$ and $180^\circ$ turns about the origin, so they can describe Earth's turning with a centre, an angle and a direction.                                                                                                                                                                                                                                                                                                                                                          |
| <b>Knowledge</b>            | Positive angle = anticlockwise turn; negative angle = clockwise turn ( $-90^\circ$ means $90^\circ$ clockwise). A $90^\circ$ angle is a quarter turn; $180^\circ$ is a half turn. About $O(0,0)$ : $90^\circ$ anticlockwise $(x,y) \rightarrow (-y,x)$ ; $-90^\circ$ ( $90^\circ$ clockwise) $(x,y) \rightarrow (y,-x)$ ; half turn $(x,y) \rightarrow (-x,-y)$ . Worked example: $A(2,1) \rightarrow A'(-1,2)$ , $B(4,1) \rightarrow B'(-1,4)$ , $C(4,3) \rightarrow C'(-3,4)$ ; $A(2,1)$ rotated $-90^\circ$ gives $A''(1,-2)$ . The angle is measured from the centre to the object point and then from the centre to the image point. |
| <b>Skills</b>               | Constructing images using tracing paper and coordinate rules; plotting points on a Cartesian plane; distinguishing clockwise and anticlockwise; using models (SEP: Developing and Using Models); analysing and interpreting data (SEP); using mathematics and computational thinking (SEP); constructing explanations (SEP).                                                                                                                                                                                                                                                                                                              |
| <b>Attitudes</b>            | Unity: learners work in pairs and groups to check each other's images. Responsibility: learners take care of rulers, tracing paper and grid paper. Appreciation of rotation in day-to-day situations. Life Skills (self-esteem): every learner contributes to the group construction.                                                                                                                                                                                                                                                                                                                                                     |
| <b>Key Inquiry Question</b> | How do we use rotation in real-life situations? Where do we apply rotational symmetry in day-to-day life?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Purpose in Storyline</b> | Lesson 2 showed that rotation keeps distances from the centre, shape and size unchanged and turns a figure through an angle in a direction. Lesson 3 located the centre using perpendicular bisectors and then measured the angle. In Lesson 4 learners use a known centre, angle and direction to construct images on a grid, including negative angles. This gives them the full 'centre, angle, direction' language that will be used in Lesson 9 to explain Mombasa, Nairobi and Kisumu day and night timing, and it prepares for the order of rotational symmetry in Lesson 5.                                                       |
| <b>Safety Notes</b>         | Keep walkways clear when learners move to the wall chart. Use blunt-tipped compass points or pencils to hold tracing paper at the centre, and handle sharp points carefully. The torch is not pointed at faces. Learners share grid paper and rulers fairly.                                                                                                                                                                                                                                                                                                                                                                              |

### B. LESSON OVERVIEW

Teacher Mwangi opens by revisiting the Driving Question Board (DQB) and the evidence so far: Earth turns about its axis (the centre), through  $360^\circ$  in 24 hours, which is  $15^\circ$  per hour, anticlockwise when viewed from above the North Pole. Learners predict where a point will land after a positive or negative turn about  $O(0,0)$ . They then observe by constructing images of triangle ABC on grid paper with tracing paper, and discover the coordinate rules for  $90^\circ$  anticlockwise,  $-90^\circ$  ( $90^\circ$  clockwise) and  $180^\circ$ . In the Explain phase, learners connect the sign of the angle to Earth's direction of turning, and see why Wanjiku in Kisumu and Otieno in Mombasa are both being carried anticlockwise even though Kenya feels still. The class updates the DQB and revises its model with a centre, an angle and a direction.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                | Learner Experience                                                 | Teacher Moves                                                    | Sensemaking Strategy                                             | Formative Assessment Strategy                                  |
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| <b>Predict Phase</b> | Learners look at the DQB and recall the time-lapse of Earth (24 s) | Point to the DQB and say: 'Last lessons we found a centre and an | Predict-Observe-Explain (POE): committing to a prediction before | Observable indicators: (1) Learner sketches show A moving in a |

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| <p><b>VIDEO:</b><br/> <a href="#">Rotating points</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a><br/> for similar videos</p> <p><b>EXERCISE:</b><br/> <a href="#">Rotate points (basic)</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a><br/> for similar readings</p>                     | <p>= 24 h, so 1 second = 1 hour = 15° of turning) with dawn over Mombasa, midday over Nairobi and darkness over Kisumu. On grid paper they plot A(2,1) with centre O(0,0). Individually they predict, with a quick sketch, where A will land after (a) a 90° anticlockwise turn and (b) a 90° clockwise turn. They write one sentence: 'I think A goes to ... because ...'. They also predict which direction Earth turns, and whether that is a positive or negative angle.</p>                                                                                                                                                                                                                                                                                                                                                             | <p>angle from an object and its image. Today we are given the centre and the angle, and we build the image.' Display the grid with A(2,1). Say: 'Wanjiku in Kisumu says Kenya feels still. If Earth turns anticlockwise, is that a positive or a negative angle? Write your prediction silently.' WAIT TIME: 15 seconds before anyone speaks. Cold-call 3 learners to share predictions for A's image without confirming them. Ask: 'What would you need to know to be sure where A lands?' Record 2 predictions on the board: one for anticlockwise, one for clockwise.</p>                                                                                                                                                   | <p>construction makes learners aware of their ideas about direction and sign, so the evidence in the Observe phase directly confirms or challenges their thinking. Linking to Earth's turning anchors the new idea in the phenomenon.</p>                                                                                                                                                                           | <p>turning path (arc around O) rather than a straight slide. (2) Learner writes the correct word, anticlockwise or clockwise, for the two turns. (3) Count how many learners link anticlockwise to positive. Note learners who place A' in the wrong quadrant for the clockwise turn, and learners who think a bigger angle means a faster Earth. Record the tally on the clipboard.</p>                        |
| <p><b>Observe Phase</b></p> <p><b>VIDEO:</b><br/> <a href="#">Rotating points</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a><br/> for similar videos</p> <p><b>EXERCISE:</b><br/> <a href="#">Rotate points</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a><br/> for similar readings</p> | <p>In pairs, learners use grid paper, tracing paper and a ruler. Task 1: Plot triangle ABC with A(2,1), B(4,1), C(4,3) and centre O(0,0). Trace the triangle and the origin, pin at O with a pencil tip, turn the tracing paper 90° anticlockwise (positive 90°), and read the image coordinates: A'(-1,2), B'(-1,4), C'(-3,4). Task 2: Turn the tracing paper 90° clockwise (-90°) and read A''(1,-2), then find B'' and C'' by the same method. Task 3: Turn through a half turn (180°). Learners complete a table of object coordinates and image coordinates for each turn, check that OA = OA' (square root of 5), OB = OB' (square root of 17), OC = OC' (5), and compare triangle sides: AB = 2, BC = 2, AC = square root of 8 equal to A'B'=2, B'C'=2, A'C'=square root of 8. They look for a pattern in how each (x,y) changes.</p> | <p>Circulate and say: 'Show me the centre first. Where does your pencil pin go?' 'Turn the paper until the line OA lies on OA'. Which way did your paper turn? Say it aloud: clockwise or anticlockwise?' For pairs who confuse direction, say: 'Think of a clock. Which way do the hands go? Anticlockwise is the opposite.' After Task 1, stop the class: 'Look at your table for A, B and C. What happens to each x and y?' WAIT TIME: 15 seconds. Cold-call 3 pairs to report what they notice. Repeat for the -90° and 180° turns. Ask: 'Why is the image of A at (1,-2) and not (-1,2)?' Hold up a prepared chart of Earth's turning: 15° per hour anticlockwise, so after 6 hours the turning is 90° anticlockwise.</p> | <p>Pattern-seeking with Analysing and Interpreting Data (SEP): tabulating object and image coordinates lets learners notice that (x,y) → (-y,x) for 90° anticlockwise, (x,y) → (y,-x) for -90°, and (x,y) → (-x,-y) for a half turn. Measuring distances connects to the Lesson 2 properties (distances from the centre and shape unchanged) and so learners test their construction rather than just trust it.</p> | <p>Observable indicators: (1) Pairs pin tracing paper at O and not at a vertex. (2) Image coordinates match A'(-1,2), B'(-1,4), C'(-3,4) and A''(1,-2). (3) Learners state OA = OA' and compare triangle sides correctly. (4) Learners use the words clockwise and anticlockwise correctly while turning. Use a checklist with a tick for each. Intervene with pairs who reverse the direction of the turn.</p> |
| <p><b>Explain Phase</b></p> <p><b>VIDEO:</b><br/> <a href="#">Determining rotations</a></p>                                                                                                                                                                                                                                                                                  | <p>Learners write the three coordinate rules in their own words and test them on a new point, such as D(3,2): they predict D' under 90°</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Say: 'Use a rule to predict, then use tracing paper to test. Who can say why the rule works for 180°?' Pose: 'Kenya feels still, but Kenya</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Claim-Evidence-Reasoning with Constructing Explanations (SEP): learners state a claim about the rule, support it with coordinates</p>                                                                                                                                                                                                                                                                            | <p>Observable indicators: (1) Learners apply a rule to D(3,2) and obtain (-2,3), (2,-3) and (-3,-2). (2) Learners say -90° means 90°</p>                                                                                                                                                                                                                                                                        |

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| <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b></p> <p><a href="#">Rotate shapes</a></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                   | <p>anticlockwise using <math>(x,y) \rightarrow (-y,x)</math>, giving <math>(-2,3)</math>, D" under <math>-90^\circ</math> using <math>(x,y) \rightarrow (y,-x)</math>, giving <math>(2,-3)</math>, and the half turn image <math>(-3,-2)</math>. They check one with tracing paper. Then in groups of four they explain, with a labelled sketch, how rotation about O is described using a centre, an angle and a direction, and connect it to Earth: centre = Earth's axis, angle = <math>15^\circ</math> per hour, direction = anticlockwise (positive). They answer: 'After 6 hours, through what angle has Kenya turned? After 12 hours?' (<math>90^\circ</math> anticlockwise and <math>180^\circ</math>). Wanjiku and Otieno are both carried along: the Sun only appears to move east to west.</p> | <p>turns with Earth. What is the centre, the angle and the direction?' WAIT TIME: 15 seconds, then pairs talk for 1 minute. Cold-call 4 learners. Press for reasons: 'How do you know the turn is positive?' 'Does the size of the triangle change?' Revoice: 'So you say the image has the same shape and size, only the position changes.' Use the sentence frame on the board: 'The rotation has centre ____, angle ____ and direction ____.' Address misconception: 'Is <math>-90^\circ</math> the same as <math>90^\circ</math> anticlockwise? Test with A.' Link: 'Mombasa sees sunrise before Kisumu because Earth turns anticlockwise. This is the same kind of turning, with <math>15^\circ</math> per hour.'</p> | <p>from the tracing-paper evidence, and reason using the properties of rotation. The sentence frame (centre, angle, direction) builds the mathematical language needed for the phenomenon.</p>                                                                                                                         | <p>clockwise. (3) Group explanations name the centre as Earth's axis, the angle as <math>15^\circ</math> per hour and the direction as anticlockwise. (4) Learners say the Sun only appears to move. Use a 3-point check: correct rule, correct sign, correct link to Earth. Note groups who still say Earth 'moves' without a centre, angle or direction.</p> |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b></p> <p><a href="#">Examples: Determine Angles of Rotation</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: CK-12</p> <p><a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b></p> <p><a href="#">Rotate points (basic)</a></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar readings</a></p> | <p>Learners return to the DQB wall chart. Each pair writes one sticky note: either a new question (for example, 'How many degrees has Kisumu turned from sunrise to sunset?') or a partial answer (for example, 'Earth turns anticlockwise, which is a positive angle'). They place notes beside the original questions and mark those now partly answered by today's rotation evidence. They also write the evidence-link: 'Today we learned that ... so ...'.</p>                                                                                                                                                                                                                                                                                                                                       | <p>Say: 'Come to the board in turns, one pair at a time. Read your note aloud and tell us which question it helps with.' Ask: 'Which of our questions can we now say more about, and which still need evidence?' WAIT TIME: 10 seconds. Cold-call 3 pairs. Group similar notes. Highlight questions to be handled next: 'Why do some shapes fit onto themselves after a turn? That is Lesson 5.' Keep the DQB neat and ensure learners move safely.</p>                                                                                                                                                                                                                                                                    | <p>Question Tracking with the Driving Question Board: connecting each new idea (positive and negative angles, rules for <math>90^\circ</math> and <math>180^\circ</math>) to the learners' own questions shows that the learning serves the puzzle of daily day and night, and keeps curiosity and ownership high.</p> | <p>Observable indicators: (1) Notes connect to rotation evidence, not random facts. (2) At least one note mentions direction or angle. (3) Learners tell which question they are answering. Count notes that use the words 'centre', 'angle' or 'direction' correctly. Identify learners whose notes show gaps to revisit.</p>                                 |
| <p><b>Model Building Phase</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Learners revisit their Lesson 1 globe-and-torch model drawing (the globe is tilted about <math>23.5^\circ</math> and</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Say: 'Take out your first model. What can we now add using today's mathematics?'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Developing and Using Models (SEP) with cumulative model revision: adding centre, angle and</p>                                                                                                                                                                                                                      | <p>Observable indicators: (1) Revised model labels the centre, angle (<math>15^\circ</math> per hour) and anticlockwise</p>                                                                                                                                                                                                                                    |

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| <p> <b>VIDEO:</b><br/> <a href="#">Counting Right Angles in a Turn</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Rotate points</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>the torch is the Sun). In groups they revise it by adding labels: a centre (Earth's axis through the North and South Poles), a direction arrow (anticlockwise as viewed from above the North Pole, a positive angle), and an angle (15° per hour, so 90° in 6 hours, 180° in 12 hours, 360° in 24 hours). They mark Mombasa, Nairobi and Kisumu on the globe and, with the torch on, turn the globe anticlockwise through 90° and describe where each place moves. They write a short caption: 'Earth's rotation can be described by centre ____, angle ____ per hour, direction ____.'</p> | <p>Demonstrate one 90° anticlockwise turn of the globe while learners observe the three cities. Ask: 'If we turned the globe clockwise, which sign would the angle have? Would the Sun appear to rise in the east?' WAIT TIME: 15 seconds. Cold-call 3 groups to explain their revised model. Prompt: 'What does your model explain now that it could not explain before? What does it still not explain?' (for example, why Mombasa has sunrise about 20 minutes before Kisumu, to be developed in Lesson 9). Keep the torch pointed away from faces.</p> | <p>direction to the globe model turns a picture into a mathematical model that can make predictions, and identifying its limits sets up later lessons.</p> | <p>direction. (2) Group states that 6 hours is a 90° anticlockwise turn and 12 hours is 180°. (3) Group identifies at least one thing the model cannot yet explain. Use a 3-item rubric (centre, angle, direction). Collect the models to check progress.</p> |
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## D. TEACHER REFLECTION

Did learners separate the sign of the angle from the direction of turn (positive = anticlockwise, negative = clockwise)? Which pairs placed the image in the wrong quadrant for -90°, and what did they say about their reasoning? Did the tracing-paper construction help them see that the distances from the centre stay the same? Did the link to Earth's turning (15° per hour, anticlockwise) feel natural or forced? Which learners still say 'Kenya is still'? What will I adjust before Lesson 5 on the order of rotational symmetry?

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                      |
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| <b>What did I observe?</b>                   | We constructed images of triangle ABC about O(0,0) using tracing paper and grid paper: 90° anticlockwise gave A'(-1,2), B'(-1,4), C'(-3,4); -90° (90° clockwise) gave A''(1,-2); a half turn changed (x,y) to (-x,-y). Distances from O and the sides of the triangle stayed the same.                                               |
| <b>What did I learn?</b>                     | A positive angle means an anticlockwise turn and a negative angle means a clockwise turn. Rules about O: 90° anticlockwise (x,y) → (-y,x); -90° (x,y) → (y,-x); 180° (x,y) → (-x,-y). Any rotation needs a centre, an angle and a direction.                                                                                         |
| <b>How does this explain the phenomenon?</b> | Earth's turning is a rotation with centre at its axis, angle 15° per hour and direction anticlockwise (positive) as seen from above the North Pole. In 6 hours it turns 90° and in 12 hours 180°. Kenya turns with Earth, and the Sun only appears to move from east to west, which helps explain the regular day and night pattern. |

## LESSON 5 (40 minutes (one double period may be used if available)): Order of Rotational Symmetry of Plane Figures: How Many Times Does a Figure Fit Onto Itself in One Full Turn?

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>Purpose</b>              | Learners count the order of rotational symmetry of plane figures (square, equilateral triangle, regular hexagon, rectangle, regular pentagon, kiondo pattern, 8-spoke bicycle wheel, matatu wheel) and find the smallest angle of fit as $360^\circ \div \text{order}$ . This advances the driving question by showing that a full turn of $360^\circ$ can be split into equal angle steps, just as Earth's $360^\circ$ turn in 24 hours splits into $15^\circ$ per hour.                                                                                                                    |
| <b>Knowledge</b>            | Order of rotational symmetry is the number of times a figure fits onto itself in one full turn of $360^\circ$ . The smallest angle of fit = $360^\circ \div \text{order}$ . Square: order 4 ( $90^\circ$ ); equilateral triangle: 3 ( $120^\circ$ ); regular hexagon: 6 ( $60^\circ$ ); rectangle (non-square): 2 ( $180^\circ$ ); regular pentagon: 5 ( $72^\circ$ ); isosceles non-equilateral triangle: 1 (no rotational symmetry); kiondo star-pattern base: 6 ( $60^\circ$ ); bicycle wheel with 8 equally spaced spokes: 8 ( $45^\circ$ ); matatu wheel with 5 nuts: 5 ( $72^\circ$ ). |
| <b>Skills</b>               | Using tracing paper and a pin to turn a figure about its centre; counting fits in a full turn; calculating $360^\circ \div \text{order}$ ; using models and mathematics (NGSS SEP: Using Mathematics and Computational Thinking; Analyzing and Interpreting Data); constructing explanations from evidence.                                                                                                                                                                                                                                                                                  |
| <b>Attitudes</b>            | Appreciating the use of rotational symmetry in day-to-day life (kiondo weaving, wheels, wind turbines). Unity when working in groups, and responsibility in handling scissors, pins and tracing paper. Curiosity about how a repeating pattern in a wheel is like the repeating pattern of day and night.                                                                                                                                                                                                                                                                                    |
| <b>Key Inquiry Question</b> | How do we use rotation in real-life situations? Where do we apply rotational symmetry in day-to-day life?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Purpose in Storyline</b> | Lessons 2 and 4 taught that a rotation turns a figure through an angle about a centre, keeping size and shape, and that a full turn is $360^\circ$ . Lesson 5 turns this into counting: how many times does a figure land exactly on itself during one $360^\circ$ turn? This prepares for Lesson 6 (axes and order of solids, including Earth's axis in the globe model) and Lesson 8 (real devices). It also links back to Lesson 1: Earth makes one full $360^\circ$ turn in 24 hours, $15^\circ$ per hour, a regular stepping of a full turn.                                            |
| <b>Safety Notes</b>         | Pins and compass points: push the pin into a soft pad or eraser board, never toward a classmate, and give them back at the end. Use blunt-nosed scissors and cut shapes only when seated. Keep walkways clear when learners pass around the models. Wheels and baskets are brought as pictures or small safe samples only; do not spin real bicycle wheels indoors.                                                                                                                                                                                                                          |

### B. LESSON OVERVIEW

Teacher Mwangi reminds learners of the day-night time-lapse: 24 hours of Earth turning in 24 seconds, Mombasa at dawn about 6:30 am, Nairobi at midday about 12:30 pm, Lake Victoria sunset about 6:40 pm, and Kisumu dark about 7:30 pm. Wanjiku in Kisumu said 'Kenya feels still', yet Kenya is turning at  $15^\circ$  per hour. Today learners ask a new question: when something turns about a centre, how often does it look exactly the same? Learners predict the order of rotational symmetry of familiar shapes and patterns, then test with tracing paper and a pin at the centre. They collect data for the square, equilateral triangle, regular hexagon, rectangle, regular pentagon, isosceles triangle, Mama Achieng's kiondo pattern (order 6), Mzee Kamau's 8-spoke bicycle wheel (order 8) and a matatu wheel with 5 nuts (order 5). They discover the rule  $\text{smallest angle} = 360^\circ \div \text{order}$  and connect it to Earth:  $360^\circ \div 24 \text{ h} = 15^\circ$  per hour. They record this on the DQB and add a rotation-symmetry piece to the class model.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Sensemaking Strategy                                                                                                                                                                                                                                                                             | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Rotation Symmetry Principles - Basic</a><br>Source: CK-12<br> <a href="#">Search ARES</a><br><a href="#">for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Rotational symmetry of 2D shapes</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a><br><a href="#">for similar readings</a>                         | <p>Learners sit in groups of four. Teacher Mwangi shows a picture of the Earth time-lapse freeze-frame and then a picture of Mama Achieng's kiondo star pattern, Mzee Kamau's 8-spoke wheel and a matatu wheel with 5 nuts. Learners individually write a prediction in their books: 'If I turn each figure about its centre through one full turn of 360°, how many times will it look exactly the same as the start? Predict for: square, equilateral triangle, regular hexagon, rectangle (non-square), kiondo pattern, bicycle wheel, matatu wheel.' They also draw a quick sketch of the figure with a dot at the centre and share their prediction with a partner. Connection: a figure that turns and looks the same again is like Kenya turning with Earth and coming back to the same view of the Sun every 24 hours.</p> | <p>Say: 'Earlier we saw Kenya turning with Earth, about a centre (the axis), through an angle, in a direction. Today, instead of asking how far we turn, we ask how many times a figure looks the same during a full turn.' Show the pictures for 30 seconds. Say: 'Write your prediction silently for 3 minutes. No talking yet.' WAIT TIME: after asking 'What does it mean to look exactly the same?', wait 10 seconds before taking answers. Then cold-call 4 learners: 'Otieno-group, what is your prediction for the square and why?' Do not say if predictions are right; say 'Thank you, we will test that with evidence.' Record 3 different predictions for the square and the rectangle on the board.</p> | <p>Predict-Observe-Explain (POE) with Think-Pair-Share: writing a prediction first exposes learners' ideas (for example that a rectangle has order 4 because it has 4 corners), so that testing them in the Observe phase creates cognitive conflict and makes the new rule more meaningful.</p> | <p>Look for: (1) every learner has written a prediction for at least 5 figures before talking; (2) learners give a reason, not just a number (e.g. 'the hexagon has 6 sides so 6'); (3) note how many predict the rectangle as 4 or 2 and how many think the isosceles triangle has 3 (common misconception: number of sides = order). Tally on the board; do not correct yet.</p>                                                                                                                                                   |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Intro to rotational symmetry</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a><br><a href="#">for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Rotational symmetry of 2D shapes</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a><br><a href="#">for similar</a> | <p>Learners work in groups of 4 with a worksheet, tracing paper, a pin, a pencil, a soft pad, and cut-out shapes (square, equilateral triangle, regular hexagon, rectangle, regular pentagon, isosceles non-equilateral triangle) plus printed pictures of the kiondo star-pattern base, the 8-spoke bicycle wheel and the matatu wheel with 5 nuts. Steps: (1) Trace the figure and mark a dot at its centre and one vertex or mark with an arrow. (2) Place the traced copy over the original and push the pin through the centre. (3) Slowly turn the tracing paper anticlockwise through one full turn</p>                                                                                                                                                                                                                     | <p>Model once with the square at the front: 'Watch the corner mark. I turn it a quarter turn... it fits. Count 1. Half turn... fits, count 2.' Remind: 'The last fit, when you come back to where you started, counts as one.' Circulate and ask each group: 'Where is your centre?' How do you know?' and 'What angle did you turn to get the first fit?' Prompt a group that gets the rectangle as 4: 'Does it fit after a quarter turn? Try again slowly.' WAIT TIME: after asking 'What do you notice about the number of fits and the smallest angle?', wait 10-15 seconds. Cold-call 5 groups to report one row of their table each</p>                                                                        | <p>Data table with pattern finding (Analyzing and Interpreting Data): organising order and angle into one class table lets learners notice that order <math>\times</math> smallest angle is always 360°, so the rule emerges from evidence, not from being told.</p>                             | <p>Observable indicators: learners place the pin at the centre, not at a corner; they turn through a full 360° and count the return to the start; their table matches the fixed values (square 4, triangle 3, hexagon 6, rectangle 2, pentagon 5, kiondo 6, bicycle wheel 8, matatu wheel 5); they can state that the isosceles triangle has order 1 (no rotational symmetry). Use a checklist: 'places centre correctly', 'counts all fits', 'records angle'. Intervene with groups who have more than two errors in the table.</p> |

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| <a href="#">readings</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>of <math>360^\circ</math>, and count each time the traced figure fits exactly onto the original, including the final return to the start. (4) Record in a table: Figure   Order (number of fits in <math>360^\circ</math>)   Smallest angle of fit. Expected data: square 4 (<math>90^\circ</math>), equilateral triangle 3 (<math>120^\circ</math>), regular hexagon 6 (<math>60^\circ</math>), rectangle 2 (<math>180^\circ</math>), regular pentagon 5 (<math>72^\circ</math>), isosceles triangle 1 (no rotational symmetry), kiondo 6 (<math>60^\circ</math>), bicycle wheel 8 (<math>45^\circ</math>), matatu wheel 5 (<math>72^\circ</math>). Learners measure the smallest angle with a protractor for at least three figures and compare it with their table. Each group shares their data on a class chart.</p>                                                                                                                                                                                                                                                                                                  | <p>(square, triangle, hexagon, rectangle, pentagon). Then call 3 more learners for kiondo, bicycle wheel and matatu wheel. Safety: remind learners to point pins down into the soft pad.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Rotation Symmetry Principles - Basic</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Rotation of Earth (Flexbook)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>Learners study the completed class table and, in groups, answer: (a) What is order <math>\times</math> smallest angle for each figure? (b) Write a rule connecting order and smallest angle. (c) Predict the order of a regular octagon and the smallest angle (order 8, <math>45^\circ</math>) and test it with calculation. (d) Explain why the rectangle has order 2 and not 4, and why a figure with order 1 has no rotational symmetry. (e) Mama Achieng's kiondo has order 6: if she turns the basket by <math>60^\circ</math>, what does she see? How many times will it look the same in <math>360^\circ</math>? (f) Link to the phenomenon: Earth turns <math>360^\circ</math> in 24 hours. If we think of 24 equal steps, what is the angle of each step? (<math>360^\circ \div 24 = 15^\circ</math> per hour.) How is it like a figure with order 24? (Note: Earth is not a pattern with 24 identical parts, but the same division of <math>360^\circ</math> applies.) Each learner writes a short claim-evidence-reasoning explanation: 'The order of rotational symmetry of a figure is ... because ...'.</p> | <p>Ask: 'Look at the last two columns. What do you get when you multiply order by smallest angle?' WAIT TIME: 10-15 seconds, then cold-call 3 learners. Press for reasoning: 'Why must it be <math>360^\circ</math>?' Bring out that if the figure fits <math>n</math> times in one full turn and the steps are equal, then <math>n \times \text{angle} = 360^\circ</math>, so <math>\text{angle} = 360^\circ \div n</math>. Write the rule on the board: 'smallest angle = <math>360^\circ \div \text{order}</math>'. Then ask: 'Mzee Kamau's wheel has 8 equally spaced spokes. What is the smallest angle? Work with your partner.' (<math>45^\circ</math>.) Cold-call 4 learners for this and the matatu wheel (5 nuts <math>\rightarrow 72^\circ</math>). Link: 'Earth turns <math>360^\circ</math> in 24 hours. Which calculation do we use to find the angle per hour?' (<math>360^\circ \div 24 = 15^\circ</math>.) Clarify carefully that Earth's day-night cycle is a turning, not a symmetrical pattern, and we use the same division idea. Address misconception: sides do not always equal order (rectangle has 4 sides but order 2).</p> | <p>Claim-Evidence-Reasoning (CER) with whole-class discussion: learners state the rule (claim), cite their table data (evidence) and explain why equal steps must add up to <math>360^\circ</math> (reasoning). This builds an argument based on data and connects the symmetry idea to the <math>15^\circ</math> per hour rate used for Earth.</p> | <p>Indicators: learners write or say 'order <math>\times</math> angle = <math>360^\circ</math>' or 'angle = <math>360^\circ \div \text{order}</math>'; they correctly calculate <math>45^\circ</math> for an 8-spoke wheel and <math>72^\circ</math> for a 5-nut matatu wheel; they explain why a rectangle (non-square) has order 2 with <math>180^\circ</math> as the smallest angle; they compute <math>360^\circ \div 24 = 15^\circ</math>. Exit-ticket style mini-check: 'A figure has order 6. What is its smallest angle? Name one Kenyan object with this order.' (<math>60^\circ</math>, kiondo). At least 80% correct signals readiness for Lesson 6; otherwise revisit with the regular hexagon.</p> |

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| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Intro to rotational symmetry</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Rotation of Earth Practice</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar readings</p>                                                                | <p>Learners revisit the Driving Question Board (wall chart) with the driving question: 'What do you think makes it possible for a daily pattern of day and night experience on Earth?' In groups they write on sticky notes (a) one thing they now know from today (e.g. 'A full turn is 360° and can be split into equal steps; Earth's 15° per hour is one such split'), and (b) one new question (e.g. 'Does a globe have rotational symmetry? How many fits?', 'Why does Mombasa get sunrise before Kisumu?', 'What is the order of a wind turbine at Lake Turkana?'). They place notes under categories: 'Rotation: centre, angle, direction', 'Order of symmetry', 'Earth and time', 'Real-life uses'. They also mark which earlier questions are partly answered.</p>                             | <p>Say: 'Look at the board. Which of our questions did today's evidence help with?' WAIT TIME: 10 seconds. Cold-call 3 groups to place a note and explain: 'What evidence from our tracing paper experiment supports this?' Point out new questions that will be answered later: the globe and solids (Lesson 6), real devices like Lake Turkana turbines (Lesson 8), and Mombasa-Kisumu timing (Lesson 9). Ask: 'Which questions can we answer now, and which need more evidence?' Move answered questions to a 'We can explain this' column and leave others open. Make sure at least one note links order of symmetry to the day-night pattern.</p>                                                                          | <p>Driving Question Board with question sorting: grouping questions and showing which are answered by evidence makes the class's thinking visible and gives purpose to the next lessons, while keeping the anchoring phenomenon at the centre.</p>                                                | <p>Indicators: each group posts at least one 'I know' note with evidence and one 'I wonder' question; at least one question connects the symmetry work to Earth's rotation; learners can point to the DQB and say which question today's data helps answer. Note vague questions (e.g. 'Why symmetry?') and coach rewriting: 'What would you measure or count to find out?'</p>                                                                           |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Identifying transformations</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Mapping shapes</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar</p> | <p>Learners return to the class model of the globe-and-torch (globe tilted about 23.5°, torch as the Sun, turned anticlockwise through 15° for each hour as seen from above the North Pole). Using a paper protractor ring placed around the globe's equator and 24 marked points, learners divide 360° into 24 equal steps of 15° and turn the globe one step at a time, noting which of Mombasa, Nairobi and Kisumu comes into light first. They compare this to a figure with order 24: each step is the same angle and the figure (or ring) looks the same after each step, though Earth's surface is not identical at every step. Learners add to their model diagrams: (1) centre = Earth's axis, (2) angle = 15° per hour, (3) direction = anticlockwise, and now (4) a note: 'A full turn of</p> | <p>Say: 'Our model has three parts: centre, angle and direction. Today we add the idea of equal steps in a full turn.' Demonstrate turning the globe 15° for 1 hour, then 90° (6 hours), then 180° (12 hours), and ask: 'How many 15° steps make a quarter turn?' WAIT TIME: 10-15 seconds before cold-calling 3 learners (answer: 6). Ask: 'If a kiondo has order 6, how many 60° steps return it to the start? How is this like Earth taking 24 steps of 15°?' Cold-call 2 more learners. Invite two learners to revise their diagrams: 'What did you add or change in your model today? What evidence made you change it?' Close with: 'In the next lesson we will explore solids and the axis that Earth turns around.'</p> | <p>Cumulative model revision (Developing and Using Models): learners add the equal-steps idea to the existing globe model, which strengthens the link between mathematical rotation (centre, angle, direction) and the physical day-night cycle, and shows how models grow as evidence grows.</p> | <p>Indicators: model diagram shows centre (axis), angle (15° per hour), direction (anticlockwise) and the new idea (equal steps; 360° ÷ number of steps); learners correctly state 6 steps of 15° make 90°; learners label a kiondo with order 6 and 60°. Ask each learner to complete: 'Today I learned that... so in our model I changed...'. Collect these as evidence of understanding and note anyone still equating order with number of sides.</p> |

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| readings | 360° can be shared into equal steps; order tells us how many.'<br>They draw one example (e.g. a kiondo with 6 steps of 60°, or the 24-step ring with 15° steps) beside the Earth model. |  |  |  |
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### D. TEACHER REFLECTION

After the lesson, review the following: (1) How many groups correctly gave the rectangle order 2 and the isosceles triangle order 1 after testing, and what helped them change their predictions? (2) Did learners keep the pin at the centre and count the return to the start as a fit? Which groups struggled with counting? (3) How well did learners link  $360^\circ \div \text{order}$  to  $360^\circ \div 24 = 15^\circ$  per hour without claiming that Earth is a symmetrical pattern? (4) Which DQB questions emerged (e.g. order of a globe, Lake Turkana turbines) and are they linked to the next lessons? (5) Were Kenyan contexts (Mama Achieng's kiondo, Mzee Kamau's bicycle wheel, matatu wheel) meaningful and did every learner participate? Plan: give extra practice using regular polygons for learners who confuse sides with order, and extension (regular octagon, 12-spoke wheel) for fast finishers.

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>What did I observe?</b>                   | We turned a square, equilateral triangle, regular hexagon, rectangle, regular pentagon, isosceles triangle, a kiondo pattern, an 8-spoke bicycle wheel and a 5-nut matatu wheel through one full turn of $360^\circ$ and counted how many times each fitted onto itself. Orders found: square 4, triangle 3, hexagon 6, rectangle 2, pentagon 5, isosceles triangle 1, kiondo 6, bicycle wheel 8, matatu wheel 5. |
| <b>What did I learn?</b>                     | The order of rotational symmetry is the number of times a figure fits onto itself in one full turn of $360^\circ$ . The smallest angle of fit = $360^\circ \div \text{order}$ (e.g. square $90^\circ$ , hexagon $60^\circ$ , 8-spoke wheel $45^\circ$ , matatu wheel $72^\circ$ ). Number of sides does not always equal the order (a rectangle has order 2). A figure with order 1 has no rotational symmetry.   |
| <b>How does this explain the phenomenon?</b> | A full turn of $360^\circ$ can be divided into equal steps. In the same way Earth turns $360^\circ$ in 24 hours, which is $360^\circ \div 24 = 15^\circ$ per hour, anticlockwise about its axis. This regular turning is part of why Kenya, from Mombasa to Nairobi to Kisumu, has a repeating daily pattern of day and night, even though Kenya feels still.                                                     |

## LESSON 6 (40 minutes (one double lesson of 2 x 40 minutes recommended if time allows)): Axis and Order of Rotational Symmetry of Solids: From Cuboids to Earth's Axis

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <b>Purpose</b>              | Learners identify axes of rotational symmetry of solids, state the order about each axis, and connect the globe's axis through the poles to the turning that gives Kenya day and night.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Knowledge</b>            | An axis of rotational symmetry of a solid is a line through the solid about which it fits onto itself in less than a full turn of $360^\circ$ . The order about an axis is the number of times the solid fits onto itself in one full turn, and the smallest angle of fit is $360^\circ \div \text{order}$ . A cuboid (non-square ends) has 3 axes, each of order 2. A square-based prism has one axis of order 4 and two axes of order 2 through opposite face centres. A cone has 1 axis of order infinity (every angle). A sphere/globe has an axis through the North and South Poles of order infinity. Extension only: a cube has 13 axes in total (3 of order 4, 4 of order 3, 6 of order 2). |
| <b>Skills</b>               | Observing and handling solid models; turning models about a chosen axis and counting fits; calculating $360^\circ \div \text{order}$ ; recording axes and orders in a table; using a model to explain a phenomenon (NGSS Practices: Developing and Using Models; Analyzing and Interpreting Data; Constructing Explanations).                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Attitudes</b>            | Curiosity about how mathematics describes the turning of familiar objects; appreciation of rotational symmetry in day-to-day life; unity and responsibility when sharing and handling models in groups.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Key Inquiry Question</b> | How do we use rotation in real-life situations? Where do we apply rotational symmetry in day-to-day life?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Purpose in Storyline</b> | Lesson 5 taught learners to count the order of rotational symmetry of plane figures and find the smallest angle as $360^\circ \div \text{order}$ . Lesson 1 gave the globe-and-torch model where Earth turns about an axis. Lesson 6 extends order from flat shapes to solids by asking about a line (an axis) as the centre of turning in 3D. It links back to the driving question by showing that the globe's axis through the poles is the centre of the rotation that gives Mombasa, Nairobi and Kisumu day and night. Lesson 7 will then show an object and its rotated image are directly congruent.                                                                                         |
| <b>Safety Notes</b>         | Handle models and the globe gently; keep fingers away from the cone's tip. Do not shine the torch into anyone's eyes. Keep walkways clear when groups gather at the globe station. Sharpened pencils used as axes must be held with the point down or away from faces.                                                                                                                                                                                                                                                                                                                                                                                                                              |

### B. LESSON OVERVIEW

Teacher Mwangi's Grade 10 class in Nairobi returns to the time-lapse snapshot: dawn over Mombasa (about 6:30 am), midday over Nairobi (about 12:30 pm), sunset over Lake Victoria (about 6:40 pm) and darkness over Kisumu (about 7:30 pm), repeating every 24 hours. Wanjiku in Kisumu says 'Kenya feels still', yet Earth turns once ( $360^\circ$ ) in 24 hours at  $15^\circ$  per hour. Learners now ask: if a solid turns about a line, how many times does it fit onto itself? They predict, then use cardboard cuboids, a square-based prism (like a box of mandazi cut as a square block), a cone (like a traditional hat or a funnel) and a globe to find axes and orders. They explain why the globe has one special axis, the line through the poles, and how it works as the 'centre' of Earth's rotation. They add to the DQB and revise their model by labelling the axis, the angle ( $15^\circ$  per hour) and the direction (anticlockwise seen from above the North Pole).

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                  | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Recognizing common 3D shapes</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br><a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Identify geometric solids (3D shapes)</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br><a href="#">Search ARES for similar readings</a> | Learners recall the day-night video and Wanjiku's statement 'Kenya feels still'. Each learner is given a cuboid (non-square ends, e.g. a cardboard matchbox or tea-leaf packet) and asked to predict individually: 'If I push a pencil through the middle of two opposite faces and turn the box, how many times will it look exactly the same in one full turn of 360°?' They sketch the box, draw the pencil as an axis, and write a prediction number and reason in their notebooks. They then predict whether the globe has an axis, and where it might be. Prior knowledge from Lesson 5 (order, 360° ÷ order) is used to predict the angle of fit. | Display the 'Mombasa dawn / Nairobi midday / Kisumu dark' snapshot. Say: 'Last time we counted how many times a flat shape fits onto itself. Today our shapes are solid, and the turning happens around a line, an axis. Earth has one too.' Hold up a cuboid and a pencil. Say: 'Write your prediction silently first.' Give 2 minutes of silent writing. Then say: 'Turn to your partner and share your number and your reason.' Cold-call 4 learners, asking: 'What number did you predict, and what is your reasoning?' Record all predictions on the board without judging. After each answer, WAIT TIME 10 seconds before calling the next learner. Ask: 'Who predicted differently? What made you choose that?' Finish with: 'Where on the globe would you put the pencil so that Mombasa, Nairobi and Kisumu all turn together?' | Predict-and-record (public prediction chart). Learners commit to an idea with a reason before evidence, which exposes prior ideas from Lesson 5 (e.g. 'a rectangle has order 2, so a box might too'). The recorded predictions are revisited after the Observe phase, so learners see their own thinking change.                                                      | Observe whether learners (1) draw the axis as a line through opposite faces, (2) link their number to 360° ÷ order, and (3) mention that the globe's axis goes from pole to pole. Indicators: at least 3 of 4 cold-called learners give a number with a reason. Note learners who confuse an axis with a mirror line (reflection) for follow-up.                                                                                                                                                                                               |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Rotation Symmetry Principles - Basic</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: CK-12<br><a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Rotational symmetry of 2D shapes</a><br><i>Related topic: not an exact match for this lesson.</i>                                                                                                                                  | In groups of 4, learners rotate through three model stations and one globe station, each with a handout table: Solid   Axis described   Order   Smallest angle of fit (360° ÷ order). Station 1: cuboid (non-square ends). Learners push a pencil through opposite face centres, turn the box slowly, count fits, and repeat for all 3 pairs of opposite faces. They find 3 axes, each of order 2 (angle 180°). Station 2: square-based prism (e.g. a block of soap or a square-ended wooden bar). They find one long axis through the two square ends with order 4 (angle 90°) and two axes through the centres of opposite rectangular                 | Say: 'You are scientists today. Do not tell me an answer, show me evidence from the model.' Give each group a role card (Turner, Counter, Recorder, Checker). Circulate and ask: 'Where is your axis? Show me with the pencil.' 'How do you know it fits exactly, not almost?' 'If the order is 4, what is the smallest angle?' At the cone station ask: 'Is there any angle where the cone does not fit onto itself?' WAIT TIME 10-15 seconds after each question before prompting. At the globe station ask: 'Which way do I turn the globe so that Mombasa meets the torch light before Kisumu?' and 'Where is Nairobi relative to the axis, and                                                                                                                                                                                      | Structured data collection with a shared table plus 'hands-on modelling'. Learners gather the same kind of evidence on different solids, which makes patterns visible: the order depends on the shape of the cross-section perpendicular to the axis. Working with the globe lets them see an axis as a centre of rotation, connecting 3D symmetry to the phenomenon. | Check the handout tables: Cuboid = 3 axes, order 2, 180°; Square prism = 1 axis order 4 (90°) and 2 axes order 2 (180°); Cone = 1 axis, order infinity; Globe = 1 axis through the poles, order infinity. Indicators: groups use the pencil correctly through face centres, state order before computing angle, and correctly name the turning direction as anticlockwise seen from above the North Pole. Tick a checklist for each group, and note common errors such as counting axes through edges or corners of a cuboid as symmetry axes. |

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| <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a><br/> <a href="#">for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>faces, each of order 2 (angle <math>180^\circ</math>). Station 3: cone (paper hat or funnel). They turn it about the line from the tip to the centre of the base and discover it fits at every angle: order infinity. Station 4: globe tilted about <math>23.5^\circ</math> with a torch as the Sun. They locate the axis through the North and South Poles, turn the globe anticlockwise (viewed from above the North Pole) through <math>15^\circ</math> for each hour, and note Mombasa, Nairobi (about <math>1^\circ</math> S) and Kisumu moving through the torch light. Fast groups attempt the extension: a cube has 13 axes (3 of order 4, 4 of order 3, 6 of order 2).</p>                                                                                                                                                                                          | <p>how far does it travel in one hour?' Rotate groups every 5 minutes with a clear signal. Monitor safe handling. Cold-call 3 groups at the end to report one finding with evidence.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Recognizing common 3D shapes</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a><br/> <a href="#">for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Rotation Symmetry (Flexbook)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: CK-12<br/> <a href="#">Search ARES</a><br/> <a href="#">for similar readings</a></p> | <p>Learners return to their predictions and compare them with evidence. In pairs, they complete the sentence frames: 'The cuboid has ____ axes, each of order ____, because ____.' 'The square-based prism has a special axis of order ____ because its ends are ____.' 'The cone and globe both have order infinity because ____.' They explain why the square prism has an axis of order 4 but the cuboid with non-square ends does not, linking this to Lesson 5 (square: order 4; rectangle: order 2). Then they use the globe model to explain in writing: Earth's axis is the centre of rotation; Earth turns <math>360^\circ</math> in 24 hours at <math>15^\circ</math> per hour, anticlockwise from above the North Pole; Mombasa turns into sunlight sooner than Kisumu. They state that Kenya is not stationary: Wanjiku and Otieno are both turning with Earth.</p> | <p>Say: 'Return to your prediction. Circle in green what you got right and in red what changed.' Then: 'Why does the square prism have order 4 along its long axis but the cuboid only order 2?' WAIT TIME 15 seconds, then pair-share for 1 minute, then cold-call 4 learners. Probe: 'What does the cross-section look like when you look along the axis?' Draw a square and a rectangle end-on, and guide learners to see that the order of the cross-section (Lesson 5) gives the order of the axis. Then ask: 'If the globe is a sphere, how many angles does it fit at? What does that tell you about the order?' Finally ask: 'Wanjiku says Kenya feels still. Using centre, angle, direction, what do you say to her?' Cold-call 3 learners and revoice answers using the words axis, order, angle, direction. Correct misconceptions gently (e.g. the Sun does not move; Earth rotates, we do not say revolution for the spin).</p> | <p>Claim-Evidence-Reasoning with sentence frames. Learners make a claim about each solid, support it with their hands-on evidence, and reason using the formula <math>360^\circ \div \text{order}</math>. Linking cross-section symmetry (Lesson 5) to solids builds transfer of knowledge rather than memorisation. Using the same words (centre, angle, direction) for Earth and for solids unifies the concept.</p> | <p>Listen for accurate use of 'axis', 'order' and '<math>360^\circ \div \text{order}</math>'. Indicators: learners correctly give <math>180^\circ</math> for order 2 and <math>90^\circ</math> for order 4; they explain infinity as 'fits at every angle'; they state Earth's rotation as centre = axis through the poles, angle = <math>15^\circ</math> per hour, direction = anticlockwise. Use a 3-question exit check on mini-whiteboards: (1) Order and angle of the cuboid axis? (2) How many axes of order 2 has a square-based prism? (3) What is the centre of Earth's rotation? Expect at least 80% correct; reteach to the rest next lesson.</p> |

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| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Rotation</a><br/> <a href="#">Symmetry</a><br/> <a href="#">Principles - Basic</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Rotation</a><br/> <a href="#">Symmetry</a><br/> <a href="#">Practice</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>Learners go to the wall DQB. They review earlier questions and identify which ones are now partly answered, for example 'What is the centre of the turning?' (the axis through the poles) and 'Why does Kenya feel still?'. They write new questions on sticky notes, such as 'Why does the globe have infinite order but a cuboid does not?', 'Does the tilt of 23.5° change the axis?', 'Is the turned image of a solid the same size as the original?' and 'How exactly does 15° per hour decide Mombasa's sunrise before Kisumu?'. They place sticky notes in categories: Axis and centre, Angle and time, Direction, Symmetry in real life.</p>                                                                                                                           | <p>Say: 'Go back to the DQB. Which question can we now answer? Which new question has appeared?' Give each learner 2 minutes to write one new question. Invite 4 learners to read their question aloud and place it on the board; ask the class: 'Which category does this belong to?' WAIT TIME 10 seconds before any learner answers. Tick questions answered today with a green dot and write the evidence in one line beside it (e.g. 'Earth's axis = line through poles; order infinity'). Point to the key inquiry questions and ask: 'Where do we see axes of rotation in Kenya? Think of a bicycle hub at Mzee Kamau's workshop or a spinning top.' Take 2 examples and leave the rest as homework questions.</p>      | <p>Public question tracking (Driving Question Board). Learners see their growing understanding and set a purpose for the next lessons. Categorising questions helps them notice connections between lessons (axis and centre, angle and time, symmetry) and keeps the phenomenon at the centre.</p>                                         | <p>Check that new questions use lesson vocabulary and show deeper thinking (for example, linking axis to time or angle to sunrise). Indicators: at least 80% of learners add or respond to a DQB note; at least 3 questions refer to the globe's axis or the 15° per hour rate. Note learners who write only recall questions and prompt them with 'How' or 'Why'.</p>                                                          |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Recognizing common 3D shapes</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Rotation</a><br/> <a href="#">Symmetry</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: math_expression<br/>  <a href="#">Search ARES</a></p>                                                                               | <p>Learners revise their initial globe-and-torch model from Lesson 1. In groups, they sketch the globe with the axis drawn as a labelled line from the North Pole to the South Pole, mark Mombasa, Nairobi and Kisumu, show the torch (Sun) lighting one half, and add an anticlockwise arrow viewed from above the North Pole. They label: centre = Earth's axis, angle = 15° per hour (360° in 24 hours), direction = anticlockwise. They add a small inset table comparing the globe with the cuboid, square prism and cone: Solid   Axis   Order. They state the symmetry link in one sentence: 'Like the cone, the globe fits onto itself at every angle about its axis, so order infinity.' They also write one remaining question to be answered in later lessons (for</p> | <p>Say: 'Take out your first model from Lesson 1. What is missing? Add what you now know.' Provide coloured pens for new additions so revisions are visible. Circulate and ask: 'Where is your axis? Show me the centre of rotation on your drawing.' 'Which way does your arrow go, and how do you know?' 'How many degrees does Kisumu turn in one hour?' WAIT TIME 10-15 seconds after each question. Cold-call 3 groups to explain their revised model in 30 seconds using the words centre, angle, direction. Ask the class to give one improvement to each group's model. Close by saying: 'Next lesson we show that a turned object and its image are exactly the same shape and size. Keep that question in mind.'</p> | <p>Cumulative model revision. Learners add the axis and order to their first model and use colour to show change in their thinking. Representing Earth's turning with centre, angle and direction ties the mathematical description of rotation to the real phenomenon and prepares them for the final explanation in Lessons 9 and 10.</p> | <p>Use a model checklist: (1) axis drawn from pole to pole, (2) anticlockwise arrow from above the North Pole, (3) 15° per hour labelled, (4) three Kenyan cities placed with Mombasa east of Kisumu, (5) a comparison table of solids with correct orders. Indicators: groups meet at least 4 of 5 items. Look for the misconception that the Sun moves around Kenya and address it by pointing to the arrow on the globe.</p> |

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| for similar readings | example how 15° per hour gives the 20-minute sunrise difference between Mombasa and Kisumu). |  |  |  |
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| D. TEACHER REFLECTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| Which learners confused axes with mirror lines or counted edges and corners as axes, and how did the pencil-through-face-centres method help them? Did the learners link the order of the cross-section (Lesson 5) to the order of the axis of a solid? Which questions on the DQB show that learners now see Earth's axis as the centre of its rotation? What adjustment (more time at the globe station, a second cuboid for each group) will improve Lesson 7 on congruence and rotation? |

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) |                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>What did I observe?</b>                                   | What we observed: a cuboid with non-square ends turned onto itself twice in a full turn about each of 3 axes; a square-based prism had one axis of order 4 and two of order 2; a cone fitted at every angle about its axis; the globe turned about its axis through the North and South Poles, anticlockwise from above the North Pole, with Mombasa meeting the light before Kisumu. |
| <b>What did I learn?</b>                                     | What we learned: an axis of rotational symmetry is a line about which a solid fits onto itself, its order is the number of fits in 360°, and the smallest angle of fit is $360^\circ \div \text{order}$ (cuboid 3 axes of order 2; square prism order 4 and order 2; cone and sphere order infinity). Earth's axis through the poles is the centre of its rotation at 15° per hour.   |
| <b>How does this explain the phenomenon?</b>                 | How this helps explain the phenomenon: Day and night repeat every 24 hours because Earth rotates once (360°) about its axis, turning anticlockwise at 15° per hour while the Sun lights one half. The axis is the centre, 15° per hour is the angle and anticlockwise is the direction; Kenya is not still but turning with Earth, so Mombasa reaches sunrise before Kisumu.          |

## LESSON 7 (40 minutes): Congruence and Rotation: Is the Image Really the Same Shape? (Why Kenya Can Turn Without Changing)

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>Purpose</b>              | Learners show that an object and its rotated image are directly congruent by measuring sides and angles, and justify the claim using evidence. This links to the phenomenon: Kenya turns with Earth through 15 degrees every hour, yet Nairobi, Mombasa and Kisumu keep their size and shape.                                                                                                                  |
| <b>Knowledge</b>            | Rotation is a direct isometry. It preserves lengths, angles, size and shape, and only the direction of facing changes. An object and its image are directly congruent. Worked example: triangle ABC with A(2,1), B(4,1), C(4,3) turned 90 degrees anticlockwise about O(0,0) gives A'(-1,2), B'(-1,4), C'(-3,4). Sides AB = 2, BC = 2, AC = square root of 8, and A'B' = 2, B'C' = 2, A'C' = square root of 8. |
| <b>Skills</b>               | Measuring and calculating lengths with a ruler and the distance idea; tracing and overlaying with tracing paper; constructing a rotated image on grid paper (building on Lessons 2 and 4); comparing measurements in a table; justifying claims with evidence (NGSS practices: analysing and interpreting data, constructing explanations, using mathematics).                                                 |
| <b>Attitudes</b>            | Learners value precise measurement and evidence-based justification, collaborate respectfully, and appreciate that mathematical turning describes real motion such as Earth's rotation. (Teacher to use the KICD competencies, values and PCIs for Sub-Strand 3.2 verbatim from the curriculum design.)                                                                                                        |
| <b>Key Inquiry Question</b> | 1. How do we use rotation in real-life situations? 2. Where do we apply rotational symmetry in day-to-day life?                                                                                                                                                                                                                                                                                                |
| <b>Purpose in Storyline</b> | Lesson 2 established that rotation keeps distances from the centre, shape and size unchanged, and Lesson 4 had learners construct images on grids. Lesson 7 now asks: how do we prove that the image is congruent to the object? This evidence supports the final explanation that Earth and Kenya turn at 15 degrees per hour without distortion. It feeds the applications in Lesson 8.                      |
| <b>Safety Notes</b>         | Keep rulers, compasses and protractors flat on desks, and use pencils with care. Do not point compass points at classmates. Keep the torch beam away from eyes during the globe link. Keep walkways clear when sharing grid paper and tracing paper.                                                                                                                                                           |

### B. LESSON OVERVIEW

Teacher Mwangi reopens the Driving Question Board with the question: what makes the daily pattern of day and night possible? Learners recall the video of dawn over Mombasa at about 6:30 am, midday over Nairobi at about 12:30 pm, sunset over Lake Victoria at about 6:40 pm and darkness over Kisumu at about 7:30 pm. Wanjiku says Kenya feels still, and Otieno says dawn comes first at Mombasa. In this lesson learners test a big idea: when something turns, does it stay the same size and shape? They predict, then measure triangle ABC and its image A'B'C' on grid paper, trace and turn it with tracing paper, and compare sides and angles in a table. They then explain why the object and image are directly congruent, and link this to Kenya turning with Earth through 15 degrees per hour without being stretched or squashed. They add evidence to the DQB and revise their model.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
|-------|--------------------|---------------|----------------------|-------------------------------|
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| <p><b>Predict Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Determining rotations</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Rigid transformations: preserved properties</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p>          | <p>Learners revisit the time-lapse snapshots (24 hours shown in 24 seconds, 1 second = 1 hour = 15 degrees). Teacher Mwangi asks: if Kenya turns 15 degrees every hour with Earth, does Kisumu get bigger, smaller or change shape? Learners then predict on a half sheet: when triangle ABC is turned 90 degrees anticlockwise about O(0,0), will the image A'B'C' have the same side lengths and angles? Each learner writes a prediction (same, bigger, smaller, or different shape) and one reason, then compares with a partner.</p>                                                                                                                                                                                                                                                                                                                                                                                                           | <p>Say: 'Last time we constructed images on a grid. Today we ask a harder question: is the image really the same triangle?' Display the DQB and point to the question about the Sun's position. Say: 'Wanjiku said Kenya feels still. If Kenya is turning, what would stay the same about Nairobi and Kisumu?' WAIT TIME 10-15 seconds, silent, before taking responses. Cold-call 4 learners to share predictions without judging. Tally the predictions on the board under same, bigger, smaller and different shape. Say: 'Keep your prediction, we will test it with measurement.'</p> | <p>Prediction with reasoning (Predict-Observe-Explain). Committing to a written prediction and a reason exposes preconceptions, such as turning changing size, and creates a need to test them. The class tally gives a shared record to return to in the Explain Phase.</p>                     | <p>Check that each learner has written a prediction with a reason, and tally the four categories. Look for learners who say the image becomes larger or smaller (misconception), and note their names for targeted questions. Indicator: at least 80 percent of learners can state what they think will stay the same and what will change.</p>              |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Congruent shapes &amp; transformations</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Congruence &amp; transformations</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>In groups of 4, learners get grid paper, a ruler, a protractor and tracing paper. Step 1: plot A(2,1), B(4,1), C(4,3) and draw triangle ABC with centre O(0,0). Step 2: construct the image by rotating 90 degrees anticlockwise using the rule (x,y) goes to (-y,x), and plot A'(-1,2), B'(-1,4), C'(-3,4). Step 3: measure sides AB, BC, AC and A'B', B'C', A'C' with a ruler, noting that AB = 2, BC = 2, AC = square root of 8 (about 2.8 units), and the image matches. Step 4: measure angles at A, B, C and at A', B', C' with a protractor (the right angle at B and B', and 45 degrees at A and C, and at A' and C'). Step 5: also measure distances OA and OA' (square root of 5), OB and OB' (square root of 17), OC and OC' (5). Step 6: trace triangle ABC on tracing paper, pin at O with a pencil tip, turn 90 degrees anticlockwise and see whether it lands exactly on A'B'C'. Learners record all measurements in a table.</p> | <p>Circulate and ask: 'What is the centre? What is the angle? Which direction?' Say: 'Measure twice. Do your numbers match for AB and A'B'?' Ask groups who finish early: 'What is the turn if the angle is -90 degrees?' and let them find A''(1,-2). Prompt groups that disagree by a small amount: 'Is that a real difference or a measuring error of about 1 millimetre?' Cold-call 3 groups to report one side pair and one angle pair. Record the class data on the board. WAIT TIME 10 seconds after asking: 'What do you notice about the two columns?'</p>                        | <p>Data table and pattern noticing (Analysing and Interpreting Data). Placing object and image measurements side by side lets learners see equal columns themselves. The tracing paper overlay gives a second, independent line of evidence, so the claim rests on both measurement and fit.</p> | <p>Observe whether learners plot points correctly, measure to the nearest millimetre, and record both the object and image columns. Indicator: the table shows AB = A'B' = 2, BC = B'C' = 2, AC = A'C' = square root of 8, angles equal, and OA = OA' etc. Ask one learner per group to point to the tracing paper overlay and say what stayed the same.</p> |

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| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Determining rotations</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Rigid transformations: preserved properties</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p>                                     | <p>Groups use their table to write a justification: 'Triangle ABC and triangle A'B'C' are directly congruent because...' with at least two pieces of evidence (all three sides equal, all three angles equal, tracing fits exactly by turning without flipping). They then discuss what changed (position and direction of facing) and what did not (lengths, angles, size and shape). Learners connect this to the phenomenon with a sentence frame: 'When Earth rotates through 15 degrees each hour, Kisumu and Mombasa _____ because rotation _____.' They revisit their predictions and mark them as supported or revised.</p> | <p>Say: 'Convince me. Don't just say they are the same; show me a number.' Model a Claim-Evidence-Reasoning frame on the board. Say: 'Why do we say directly congruent and not just congruent?' Guide: the image is turned, not flipped. WAIT TIME 15 seconds. Cold-call 4 learners, each to give one piece of evidence. Ask: 'Would the sides change if we turned by 15 degrees, or 180 degrees, or 360 degrees?' Say: 'Mama Achieng turns her kiondo while weaving, and the pattern does not stretch. Why?' Clarify that Earth turning does not change Nairobi's size, so Kenya is not distorted even though it is moving. Do not re-teach finding the centre (Lesson 3) or symmetry (Lessons 5 and 6).</p> | <p>Claim-Evidence-Reasoning (CER) with peer critique. Learners must connect a claim to measured evidence and a reason (rotation is a direct isometry), which turns observation into explanation. Peer critique of justifications strengthens the precision of the mathematical language.</p> | <p>Collect justifications and look for: (1) a clear claim, (2) at least two measurements as evidence, (3) a reason using the words rotation, preserves, directly congruent. Indicator: learners say congruence is shown by equal sides and angles, not by 'looks the same'. Use a thumbs check (up, sideways, down) on the question: can I justify congruence with measurements?</p> |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Congruent shapes &amp; transformations</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Congruence &amp; transformations</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Learners write new notes on sticky notes and add them to the wall chart: what they found (e.g. 'Rotation keeps lengths: <math>AB = A'B' = 2'</math>'), what questions are answered (e.g. 'Does turning change shape?') and new questions (e.g. 'How can a spinning wheel use symmetry?', 'Why does Mombasa get sunrise before Kisumu if everything turns together?'). They place notes beside the question about what makes day and night possible and group them under headings such as 'centre', 'angle', 'direction', 'congruence'.</p>                                                                                       | <p>Say: 'What can we now say about Kenya turning with Earth?' Point to the DQB. Ask each group to bring one sticky note. WAIT TIME 10 seconds before the first note is shared. Cold-call 3 groups to read their notes aloud and explain where they belong. Cross off or star questions that the lesson partly answered, and remind learners that Mombasa sunrise about 20 minutes before Kisumu will be returned to in Lesson 9. Link the new questions to Lesson 8 on real devices.</p>                                                                                                                                                                                                                      | <p>Question tracking with a public record (DQB). Sorting notes by theme shows how each lesson adds evidence, keeps learners motivated by their own questions, and shows the path from a puzzling phenomenon to a mathematical explanation.</p>                                               | <p>Check that notes include evidence (a measurement) and a link to centre, angle, direction or congruence. Indicator: at least one new question per group that connects congruence to a real-life use, and notes placed under the right heading.</p>                                                                                                                                 |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Learners revise the globe-and-torch model in their notebooks. They draw Earth with its axis (the centre of rotation), an arrow</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Say: 'Add one new idea to your model from today. Make sure it includes the centre, the angle and the direction.' Circulate with the</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Cumulative model revision. Adding the idea that turning preserves shape to the existing centre-angle-direction model builds a more</p>                                                                                                                                                    | <p>Check models for an axis labelled as centre, an anticlockwise arrow, 15 degrees per hour and the added congruence statement.</p>                                                                                                                                                                                                                                                  |

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| <p><a href="#">Rotating points</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Determine rotations</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a> for similar readings</p> | <p>showing anticlockwise turning viewed from above the North Pole, and 15 degrees per hour. They add a new labelled note: 'Turning keeps size and shape: Kenya is carried around without being stretched. Object and image are directly congruent.' They sketch triangle ABC and A'B'C' beside the model with the matching sides labelled 2, 2 and square root of 8. Pairs use the globe and a torch to show Nairobi turning through 15 degrees in one hour, then say what changed (position and facing) and what did not (distances between cities).</p> | <p>globe and torch and say: 'Show me Nairobi after one hour. How many degrees did it turn? Did the distance between Nairobi and Kisumu change?' WAIT TIME 10-15 seconds. Cold-call 3 pairs to demonstrate. Prompt: 'What does the model still not explain?' Point to the Mombasa-Kisumu timing as something to return to. Ensure the torch is not pointed at faces.</p> | <p>complete explanation, and drawing both the Earth model and the triangle model links mathematics to the phenomenon.</p> | <p>Indicator: learners explain orally that Kenya turns with Earth without changing shape, using the words directly congruent or preserved.</p> |
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#### D. TEACHER REFLECTION

Which learners relied on 'looks the same' rather than measurements, and how did the tracing paper overlay change their thinking? Did any learners confuse directly congruent with reflections or enlargements? Were measuring errors of about a millimetre treated as evidence or as disagreement? How well did learners connect the triangle result to Kenya turning with Earth? What adjustments are needed before Lesson 8 on real-life applications?

#### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                           |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What did I observe?</b>                   | We observed that triangle ABC with A(2,1), B(4,1), C(4,3) and its 90 degree anticlockwise image A'(-1,2), B'(-1,4), C'(-3,4) have equal sides (2, 2, square root of 8) and equal angles, and the traced triangle fits exactly after turning.                              |
| <b>What did I learn?</b>                     | We learned that rotation preserves lengths, angles, size and shape, so the object and image are directly congruent. Only the position and direction of facing change.                                                                                                     |
| <b>How does this explain the phenomenon?</b> | We can explain that Kenya and its cities turn with Earth at 15 degrees per hour anticlockwise without being stretched or squashed, because rotation is a direct isometry, so Nairobi, Mombasa and Kisumu keep their size and shape while day and night sweep across them. |

## LESSON 8 (40 minutes (one double lesson of 80 minutes may be used if available)): Applications in Real-Life Situations: Wheels, Turbines, Clocks and Kiondo Patterns

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>Purpose</b>              | Learners apply rotation and rotational symmetry to analyse real devices and crafts (a clock, an 8-spoke bicycle wheel, the 3-blade Lake Turkana wind turbine and a 6-fold kiondo pattern) so that they can describe any turning, including Earth's turning that gives day and night over Mombasa, Nairobi and Kisumu, using a centre, an angle and a direction.                                                                                                                                                                                                                                                                                                                                                              |
| <b>Knowledge</b>            | Learners will know that: (1) the minute hand turns $360^\circ$ per 60 min = $6^\circ$ per minute and the hour hand turns $360^\circ$ per 12 h = $30^\circ$ per hour, both clockwise (negative angles); (2) the Lake Turkana turbine (Marsabit County) has 3 blades, order 3, smallest angle of fit $120^\circ$ ; (3) the 8-spoke bicycle wheel of Mzee Kamau has order 8, angle $45^\circ$ ; (4) Mama Achieng's kiondo star-pattern base has order 6, angle $60^\circ$ ; (5) smallest angle of fit = $360^\circ \div \text{order}$ ; (6) Earth turns $360^\circ$ in 24 h = $15^\circ$ per hour anticlockwise seen from above the North Pole, so each of these devices and Earth is described by centre, angle and direction. |
| <b>Skills</b>               | Science and Engineering Practices: Developing and Using Models (represent a device as a centre, angle and direction); Analysing and Interpreting Data (compute angles from time and order); Using Mathematics and Computational Thinking ( $360^\circ \div \text{order}$ , rate $\times$ time); Constructing Explanations (link device turning to Earth's turning); Engaging in Argument from Evidence. Core competencies: Critical Thinking and Problem Solving, Communication and Collaboration, Creativity and Imagination.                                                                                                                                                                                               |
| <b>Attitudes</b>            | Learners appreciate the use of rotation and rotational symmetry in day-to-day life, show unity and respect when working in groups, and take responsibility for care of the globe, torch, tracing paper and clock models. They show curiosity about Kenyan crafts and engineering.                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Key Inquiry Question</b> | 1. How do we use rotation in real-life situations? 2. Where do we apply rotational symmetry in day-to-day life?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Purpose in Storyline</b> | Lessons 5, 6 and 7 gave learners the order of rotational symmetry, axes of solids and direct congruence. In Lesson 8 they use these tools on real Kenyan devices and crafts. The clock gives a rate-times-time pattern ( $6^\circ$ per minute, $30^\circ$ per hour) that learners transfer to Earth's $15^\circ$ per hour, preparing for Lesson 9 where the globe model is revised to explain why Mombasa gets sunrise about 20 minutes before Kisumu.                                                                                                                                                                                                                                                                       |
| <b>Safety Notes</b>         | Use a torch only at arm's length and never shine it into eyes. Keep the globe on a stable table so it does not roll off. Use blunt compasses or pencils with caution. Do not look directly at the Sun when discussing it. Keep bags and walkways clear during station rotation.                                                                                                                                                                                                                                                                                                                                                                                                                                              |

### B. LESSON OVERVIEW

Teacher Mwangi reopens the Day and Night video of Mombasa dawn (about 6:30 am), Nairobi midday (about 12:30 pm), Lake Victoria sunset (about 6:40 pm) and Kisumu darkness (about 7:30 pm). Wanjiku says 'Kenya feels still', so the class asks whether the turning can be described the same way as turning things they know. Learners predict the angles and orders of four devices, then rotate through four stations: (1) a clock with a  $6^\circ$  per minute minute hand and  $30^\circ$  per hour hour hand, (2) Mzee Kamau's 8-spoke wheel, (3) the 3-blade Lake Turkana Wind Power turbine, and (4) Mama Achieng's 6-fold kiondo pattern. At every station they name the centre, angle and direction and use  $360^\circ \div \text{order}$ . In the Explain phase they compare with Earth ( $15^\circ$  per hour anticlockwise) and see that all of these are the same kind of turning. They add questions to the DQB and revise the class model with a 'rate times time' idea.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Sensemaking Strategy                                                                                                                                                                                                                                                                                                        | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Determining rotations</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>HTML:</b><br><a href="#">Angles of Rotation in Standard Positions: Clock Angles (PLIX)</a><br>Source: CK-12<br> <a href="#">Search ARES for similar readings</a>         | Learners recall the day-night video (Mombasa dawn about 6:30 am, Nairobi midday about 12:30 pm, Lake Victoria sunset about 6:40 pm, Kisumu dark about 7:30 pm). On a prediction sheet they individually write answers to: (a) In 20 minutes, how many degrees does the minute hand turn? (b) What angle must a 3-blade turbine turn to look exactly the same? (c) How many times does an 8-spoke wheel fit onto itself in one full turn? (d) Could Earth's turning be described like a clock hand? Why or why not? They then share predictions with a partner and sketch one device with a centre marked.                                                                                                                                                                                                                                                | Project the video snapshot for 1 minute and say: 'Wanjiku says Kenya feels still. If Kenya is turning, what other turning things do you know?' Hold up a toy wheel or a clock. Give 2 minutes of silent writing, then WAIT TIME of 10 seconds after each question before calling on anyone. Cold-call 4 learners (not volunteers) to share a prediction without saying right or wrong: 'Thank you, what makes you think that?' Record two or three different predictions on the board under 'Our predictions'. Circulate and note who guesses angles by measuring and who uses ideas like $360^\circ \div \text{order}$ .                                                                      | Predict-Observe-Explain (prediction first): committing to a prediction makes learners' existing ideas about angle, order and Earth's turning visible, so later evidence at the stations can confirm or challenge them. Think-Pair-Share lets quieter learners rehearse ideas before whole-class talk.                       | Look for: learners who write 'a full turn is $360^\circ$ ' or 'order is the number of times it fits'; learners who mix up order and angle; learners who link the clock to Earth. Tally on a clipboard how many predicted $120^\circ$ for the turbine and $45^\circ$ for the wheel. Indicator of readiness: at least 4 of 6 pairs state a reason (not just a number) for one prediction.                                                                                                                                                                                   |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Rotating points</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Rotational symmetry of 2D shapes</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | In groups of 4, learners rotate through four stations (about 6 minutes each), recording centre, angle, direction and order in a table. Station 1 Clock: a model clock with movable hands. Learners turn the minute hand for 10, 20 and 45 min and the hour hand for 1, 3 and 5 h, recording angles at $6^\circ$ per minute and $30^\circ$ per hour, and noting the direction is clockwise, so the angles are written as negative (e.g. 20 min gives $-120^\circ$ ). Station 2 Mzee Kamau's 8-spoke wheel: a picture or real bicycle wheel; learners count 8 equal positions, calculate $360^\circ \div 8 = 45^\circ$ , and mark the centre at the hub. Station 3 Lake Turkana Wind Power turbine (Marsabit County): a 3-blade cut-out on tracing paper; learners rotate the tracing through $120^\circ$ , $240^\circ$ , $360^\circ$ and confirm order 3. | Before starting, say: 'At each station, answer three questions first: where is the centre, how big is the angle, which direction?' Place one-sentence prompt cards at each station. Circulate and ask: 'Show me the centre. How do you know it does not move?' and 'What would happen to the order if one blade were missing?' Give WAIT TIME of 10 to 15 seconds before prompting any group. At the clock station ask: 'The minute hand turns $6^\circ$ each minute. How many degrees in 20 minutes?' Cold-call 3 groups at the end to report one surprising finding. Ring a bell every 6 minutes to rotate stations. Check that learners handle tracing paper and the clock model with care. | Data table and station-based investigation: collecting the same three descriptors (centre, angle, direction) at different devices lets learners notice the pattern across contexts. Recording order and angle side by side makes the relationship $\text{angle} = 360^\circ \div \text{order}$ visible from their own data. | Observable indicators: learners physically place a pencil on the centre before turning; table shows $6^\circ$ , $30^\circ$ , $120^\circ$ , $45^\circ$ , $60^\circ$ correctly; learners write clockwise turns as negative angles; at least 3 of 4 groups give order 3 for the turbine and order 6 for the kiondo. Teacher checklist: tick each group when they correctly justify an order by showing the tracing paper fit, not just by counting. Note groups that wrongly give 8 for the wheel by counting spokes twice or give $120^\circ$ as the order for the turbine. |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Station 4 Mama Achieng's kiondo star pattern from Kisumu: learners trace the pattern, rotate it, and count 6 fits, so order 6 and angle 60°. At each station they check that a point and its image stay the same distance from the centre, and they confirm the turned pattern is directly congruent to the original (Lesson 7).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Explain Phase</b><br> <b>VIDEO:</b><br><a href="#">Examples: Determine Angles of Rotation</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: CK-12<br> <a href="#">Search ARES for similar videos</a><br><br> <b>HTML:</b><br><a href="#">Angles of Rotation in Standard Positions: Clock Angles (PLIX)</a><br>Source: CK-12<br> <a href="#">Search ARES for similar readings</a> | Learners return to their desks and write a claim-evidence-reasoning statement answering: 'How are the clock, wheel, turbine and kiondo like Earth turning over Kenya?' They must include the three descriptors for each device. They then solve a transfer task: 'If the minute hand turns 6° per minute, and Earth turns 15° per hour, how many degrees does Earth turn in 2 hours and how many degrees does the hour hand turn in 2 hours?' (Earth: 30°; hour hand: 60°.) Finally, they link to Mombasa and Kisumu: Mombasa is ahead of Kisumu in longitude by 4.9°, and $4.9^\circ \div 15$ is about 0.33 h, about 20 min, so the same rate idea applies. They note that Earth turns anticlockwise viewed from above the North Pole, but the clock turns clockwise, which is why the clock uses negative angles. Learners also state that the wheel's 45° and the turbine's 120° are smallest angles of fit, while the clock and Earth turn through any angle. | Say: 'Choose one device and tell me its centre, angle, direction. Then tell me how it is like Earth.' Give 2 minutes writing time followed by WAIT TIME of 15 seconds before calling on learners. Cold-call 5 learners, asking each to build on the previous learner's idea: 'Who can add to Otieno's point about dawn coming first at Mombasa?' Ask probing questions: 'Is the Sun moving or are we?' and 'Which devices have order and which turn continuously?' Draw a three-column comparison on the board (Device, Centre/Angle/Direction, Link to Earth) and fill it using learners' words. Correct the misconception that order equals angle by asking learners to compute $360^\circ \div 3$ on the board. Remind learners to use 'rotation' not 'revolution' for Earth's spin. | Claim-Evidence-Reasoning with analogy mapping: learners state a claim, support it with data from the stations, and reason using the idea that turning is described the same way everywhere. Mapping the clock's rate (6° per minute) to Earth's rate (15° per hour) builds understanding of rate times time as a general tool, which supports explaining Mombasa's earlier sunrise. | Collect 6 written CER statements. Success criteria: names centre, angle and direction for at least 3 devices; correctly states 15° per hour for Earth; correctly computes 30° for 2 hours of Earth and 60° for the hour hand in 2 hours; links the Sun's apparent east to west motion to Earth turning anticlockwise. Indicator of misunderstanding: saying the Sun moves around a still Kenya or confusing order 3 with 3°. Use a thumbs-up or thumbs-down check: 'Is 120° the order of the turbine?' and note how many learners respond correctly. |
| <b>Driving Question Board (DQB) Creation</b><br> <b>VIDEO:</b><br><a href="#">Determining rotations</a><br><i>Related topic: not</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Learners revisit the wall chart DQB. In groups they write on sticky notes: (1) one question answered by today's evidence (for example, 'Does Earth turn at a steady rate like a clock hand?'), (2) one new question (for example, 'Why does Mombasa see the Sun                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Say: 'Look at the board. Which question can we now answer using the turbine, wheel, clock or kiondo?' Give WAIT TIME of 10 seconds. Invite 3 learners (cold-called) to place a note and explain it in one sentence. Ask: 'What do we still not know about why                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Driving Question Board: sorting and linking questions makes the growth of class understanding visible and gives learners ownership of what to investigate next. Categorising questions as answered, partly answered or open helps learners judge the                                                                                                                                | Indicators: every group posts at least one new question; at least 3 notes cite evidence from today's stations (not just opinions); at least one question is about timing or longitude differences, showing readiness for Lesson 9. Teacher notes which questions show                                                                                                                                                                                                                                                                                |

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| <p><i>an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b></p> <p><a href="#">Rotational symmetry of 2D shapes</a></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                        | <p>before Kisumu if all of Kenya turns together?'). They post the notes and move earlier questions into categories: answered, partly answered, still open. They add a note of evidence from today, such as 'clock: 6° per minute, steady rate; Earth: 15° per hour'.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>Mombasa gets sunrise before Kisumu?' Highlight at least two open questions that Lesson 9 will address. Praise careful questions and avoid giving the answer immediately. Ensure every group posts at least one note.</p>                                                                                                                                                                                                                                                                                                                                 | <p>strength of the evidence gathered.</p>                                                                                                                                                                                                                             | <p>confusion about the Sun moving versus Earth turning for follow-up.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b></p> <p><a href="#">Rotating points</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b></p> <p><a href="#">Determine rotations</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar readings</a></p> | <p>Learners revise their Lesson 1 globe-and-torch model on a poster or in their notebooks. They draw Earth's axis as the centre of rotation, mark an anticlockwise arrow viewed from above the North Pole, and label the rate 15° per hour and 360° in 24 h. They add labels for Mombasa, Nairobi and Kisumu at the correct east-to-west order and add a small clock inset showing 6° per minute to show that the same 'centre, angle, direction' idea works at every scale. In pairs, they use a globe tilted about 23.5° and a torch to turn the globe through 15° for each hour, observing that Mombasa moves into the light before Kisumu. They complete a sentence frame: 'Our model now shows that...'</p> | <p>Say: 'Add one new part to your model from today and explain why you added it.' Show a quick demonstration: turn the globe anticlockwise through 15° and ask 'Which city enters the light first?' Give WAIT TIME of 15 seconds. Cold-call 3 pairs to explain their change. Ask: 'What is the centre of Earth's rotation? What is the angle per hour? What is the direction?' Insist on the vocabulary 'rotation'. Collect the sentence frames. Prompt pairs that only add decoration: 'How does this change help someone else explain day and night?'</p> | <p>Cumulative model revision: learners improve the same model after each new set of evidence so the model becomes an explanatory tool, not just a picture. Using a physical globe and torch with a numeric rate links mathematical description to the phenomenon.</p> | <p>Model checklist: (1) axis shown as centre, (2) anticlockwise direction shown, (3) 15° per hour labelled, (4) 360° in 24 h stated, (5) Mombasa, Nairobi, Kisumu placed east to west, (6) a link to today's devices. Aim for 4 of 6 items from at least 80% of pairs. Exit question on a slip: 'How many degrees does Earth turn from 6:30 am to 12:30 pm?' (Answer: <math>6 \text{ h} \times 15^\circ = 90^\circ</math>.) Use slips to sort learners into those who are secure and those needing support.</p> |

## D. TEACHER REFLECTION

Which stations let learners state centre, angle and direction most confidently, and which still needed prompts? Did learners confuse order with angle or treat the clock's continuous turning as having an order? How many learners moved from saying 'Kenya is still' to 'Kenya turns with Earth at 15° per hour'? Which DQB questions should drive Lesson 9, in particular the 20-minute difference between Mombasa and Kisumu? What materials (tracing paper, clock models) need to be prepared or replaced?

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What did I observe?</b>                   | We observed that the minute hand turns $6^\circ$ per minute and the hour hand $30^\circ$ per hour (clockwise, so negative angles), the Lake Turkana turbine has order 3 ( $120^\circ$ ), Mzee Kamau's 8-spoke wheel has order 8 ( $45^\circ$ ), and Mama Achieng's kiondo pattern has order 6 ( $60^\circ$ ).  |
| <b>What did I learn?</b>                     | We learned that any turning can be described by a centre, an angle and a direction, that the smallest angle of fit is $360^\circ \div \text{order}$ , and that $\text{rate} \times \text{time}$ gives the angle turned, just as Earth turns $15^\circ$ per hour anticlockwise.                                 |
| <b>How does this explain the phenomenon?</b> | We can explain that the day and night pattern over Mombasa, Nairobi and Kisumu fits the same kind of turning as a clock or wheel: Earth rotates $360^\circ$ in 24 h about its axis at $15^\circ$ per hour, so places further east like Mombasa turn into sunlight sooner than places further west like Kisumu. |

## LESSON 9 (40 minutes (one lesson; extend to a double lesson if time is short)): Revisiting the Phenomenon: Explaining Day and Night over Mombasa, Nairobi and Kisumu with Rotation

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <b>Purpose</b>              | Learners revisit the day-night phenomenon and revise their first globe-and-torch model. They use the three parts of a rotation (centre, angle, direction) and the rate of $15^\circ$ per hour to explain why Mombasa, Nairobi and Kisumu experience sunrise, midday, sunset and darkness at different times.                                                                                                                                                                                                                                                                                                                       |
| <b>Knowledge</b>            | Earth rotates once ( $360^\circ$ ) about its axis in 24 hours, so it turns $360^\circ \div 24 \text{ h} = 15^\circ$ per hour, anticlockwise when viewed from above the North Pole. The centre of the turning is Earth's axis through the North and South Poles. Mombasa (about $39.7^\circ \text{ E}$ ), Nairobi (about $36.8^\circ \text{ E}$ ) and Kisumu (about $34.8^\circ \text{ E}$ ) differ in longitude by $2.9^\circ$ , $2.0^\circ$ and $4.9^\circ$ . Dividing each difference by 15 gives about 12, 8 and 20 minutes, so Mombasa sees sunrise about 20 minutes before Kisumu. The Sun only appears to move east to west. |
| <b>Skills</b>               | Revising and drawing a model (SEP: Developing and Using Models); calculating with rates and angles (SEP: Using Mathematics and Computational Thinking); constructing an explanation with evidence (SEP: Constructing Explanations); communicating in rotation vocabulary (centre, angle, direction).                                                                                                                                                                                                                                                                                                                               |
| <b>Attitudes</b>            | Appreciate that mathematics explains everyday events in Kenya. Show unity and respect when peers revise each other's models. Show responsibility when handling the globe and torch.                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Key Inquiry Question</b> | How do we use rotation in real-life situations? Where do we apply rotational symmetry in day-to-day life?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Purpose in Storyline</b> | Lesson 1 launched the phenomenon and built the first globe-and-torch model. Lesson 3 gave learners the idea of an angle of rotation about a centre, linked to $15^\circ$ per hour. Lesson 6 gave them Earth's axis through the poles in the globe model. In Lesson 9 learners bring these together to revise the model and explain the timing differences between three Kenyan cities. This prepares the final explanation and performance task in Lesson 10.                                                                                                                                                                      |
| <b>Safety Notes</b>         | Use a torch only. Never look directly at the Sun or shine the torch into anyone's eyes. Keep the globe on a stable table and turn it gently, only in the direction the teacher shows. Walk carefully around the model station. Handle the globe with both hands.                                                                                                                                                                                                                                                                                                                                                                   |

### B. LESSON OVERVIEW

Teacher Mwangi replays the time-lapse of Earth in which 24 hours are shown in 24 seconds, so 1 second equals 1 hour, which is  $15^\circ$  of turning. Learners first predict what will change in their Lesson 1 models. They then gather evidence with the globe (tilted about  $23.5^\circ$ ) and torch, turning it anticlockwise  $15^\circ$  for each hour and marking Mombasa, Nairobi and Kisumu. They use the longitude differences ( $2.9^\circ$ ,  $2.0^\circ$ ,  $4.9^\circ$ ) to calculate time differences of about 12, 8 and 20 minutes. In the Explain phase they write a claim-evidence-reasoning explanation of why Mombasa sees sunrise about 20 minutes before Kisumu. They then update the Driving Question Board and revise the class model with centre (Earth's axis), angle ( $15^\circ$  per hour) and direction (anticlockwise). Wanjiku in Kisumu says 'Kenya feels still', and the lesson helps learners see that Kenya is turning with Earth.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| <p><b>Predict Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Determining rotations</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Determine rotations (basic)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners look at the initial model they drew in Lesson 1 and watch the first 10 seconds of the time-lapse again (dawn Mombasa about 6:30 am, midday Nairobi about 12:30 pm, sunset Lake Victoria about 6:40 pm, darkness over Kisumu about 7:30 pm). In pairs, they predict on a sticky note: 'If Earth turns 15° each hour, how much will Mombasa have turned by the time Nairobi has its midday? Which of the cities sees sunrise first and by about how many minutes?' They also write one thing in their Lesson 1 model they think is missing: centre, angle or direction.</p>                                                                                                                                                                                                                         | <p>Say: 'Open your Lesson 1 model. Wanjiku in Kisumu says Kenya feels still. Otieno in Mombasa sees dawn first. What do you think is happening, and what is missing from your first model?' Give WAIT TIME of 10-15 seconds in silence before anyone writes. Then ask learners to think-pair for 2 minutes. Cold-call 4 pairs (not volunteers) to share one prediction each. Record all predictions on the board without judging them as right or wrong. Ask: 'Which word would help us describe the turning exactly: centre, angle or direction?'</p>                                                                                                                                                                                                                            | <p>Prediction with a revisit of the initial model (Elicit prior ideas). Learners commit to a prediction in writing so they can later compare it with evidence. This builds understanding because they see their own thinking change and recognise which parts of the model need revision.</p>                                             | <p>Observe whether learners (1) mention turning of Earth rather than the Sun moving, (2) use any of the words centre, angle or direction, and (3) attempt a number for the angle per hour or time difference. Look for sticky notes that show the misconception 'the Sun goes around Kenya' or 'Kenya is still' and note these learners for follow-up questions in the Observe phase.</p>                                           |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Rotating points</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Determine rotations</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p>         | <p>In groups of 4, learners use a globe tilted about 23.5° and a torch as the Sun in a dimmed corner. One learner holds the torch, one turns the globe anticlockwise (viewed from above the North Pole), one marks Mombasa, Nairobi and Kisumu with small sticky dots, and one records the observations. Step 1: Start with Mombasa at dawn (the boundary of light and dark just reaching it). Step 2: Turn the globe 15° (mark the 15° on a paper protractor strip or on a card placed under the globe) and record what happens at Nairobi and Kisumu. Step 3: Continue in 15° steps for a total of 6 steps and record when each dot enters light and leaves light. Step 4: Use the longitude table (Mombasa about 39.7° E, Nairobi about 36.8° E, Kisumu about 34.8° E) to record the differences 2.9°,</p> | <p>Circulate and ask: 'Where is the centre of your turning? Show me with your finger.' (Earth's axis through the poles.) 'Which way are you turning the globe, and is that a positive or negative angle?' (Anticlockwise, positive.) 'If the globe turns 15°, how many hours of Earth time is that?' (1 hour.) Give WAIT TIME of 10-15 seconds after each question. Cold-call 3 groups to demonstrate the first 15° step to the class. Remind groups: 'Turn slowly and keep the torch still and level; the Sun does not move in this model.' Then ask: 'Why does the dot for Mombasa reach the light before the dot for Kisumu?' Point out the longitude table and ask groups to complete the time calculations, checking that they have used 15° per hour and 24 h as exact.</p> | <p>Modelling with a physical model and data collection (Developing and Using Models). Learners manipulate the globe and torch while linking each turn to an angle and a time. This builds understanding because the abstract idea of 15° per hour becomes a visible, repeatable action and the numerical results match what they see.</p> | <p>Check that groups (1) turn the globe anticlockwise when viewed from above the North Pole, (2) place the torch fixed while only the globe moves, (3) correctly calculate about 12, 8 and 20 minutes from the angles, and (4) record that Mombasa enters light first, then Nairobi, then Kisumu. Use a quick checklist with 4 ticks per group. Ask any group with errors to repeat one 15° step and explain it to a neighbour.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>2.0° and 4.9°. Learners then calculate the time for Earth to turn through each: <math>2.9^\circ \div 15</math> is about 0.19 h, about 12 min; <math>2.0^\circ \div 15</math> is about 0.13 h, about 8 min; <math>4.9^\circ \div 15</math> is about 0.33 h, about 20 min. They record that places further east turn into the sunlight sooner.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Examples: Determine Angles of Rotation</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Determine rotations (basic)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Individually, then in pairs, learners write a claim-evidence-reasoning (CER) explanation for the question: 'Why does Mombasa get sunrise about 20 minutes before Kisumu, even though both cities are in Kenya?' Claim: Mombasa sees sunrise about 20 minutes earlier because Earth rotates anticlockwise and Mombasa is further east. Evidence: Mombasa about 39.7° E and Kisumu about 34.8° E give a difference of 4.9°; the globe model showed the Mombasa dot entering the light first. Reasoning: Earth turns 360° in 24 hours, which is 15° per hour, so <math>4.9^\circ \div 15</math> is about 0.33 h, about 20 minutes. Learners also describe the rotation using centre, angle and direction: centre = Earth's axis, angle = 15° per hour, direction = anticlockwise. They then use their explanation to answer Wanjiku's comment 'Kenya feels still' by explaining that the Sun only appears to move east to west because Kenya is turning with Earth.</p> | <p>Say: 'Use the three words centre, angle and direction in your explanation. Do not tell me that the Sun moves; tell me what Earth does.' Give WAIT TIME of 10-15 seconds before pairs begin to talk. Cold-call 5 learners to read one sentence each from their CER, then ask the class: 'Did the evidence support the claim? Which number was the most important and why?' Probe: 'If Earth turned 30° each hour instead of 15°, what would the day length be?' (12 hours, so this fits no real data and shows that 24 hours and 15° per hour belong together.) Use the sentence starter on the board: 'Mombasa sees sunrise before Kisumu because ... so ...'. Address misconceptions: 'Is the Sun really moving from east to west?' Refer to the established fact that the Sun only appears to move, and that the Earth rotates from west to east.</p> | <p>Claim-Evidence-Reasoning with peer critique (Constructing Explanations). Learners connect their model observations and calculations to a general principle. This builds understanding because they must justify each claim with numbers and a mechanism, not only describe what happened.</p>          | <p>Look for explanations that (1) state that Earth rotates anticlockwise at 15° per hour, (2) use the longitude difference 4.9° and calculate about 20 minutes, (3) name the centre as Earth's axis, and (4) explain that the Sun only appears to move. Use a 3-level rubric on a half sheet: Level 1 = describes the observation only; Level 2 = uses the angle or time but not both; Level 3 = uses centre, angle, direction and the time difference correctly. Aim for most learners to reach Level 2 or 3 before moving on.</p> |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Determining rotations</a><br/> <i>Related topic: not an exact match</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Learners go to the DQB wall chart with sticky notes. For each existing question (for example, 'What keeps the day-night cycle regular?', 'Why does Mombasa get sunrise before Kisumu?', 'How can the turning be described mathematically?'), they add an answer on a green note with</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Say: 'Go back to the questions we asked in Lesson 1. Which can we now answer with evidence?' Give WAIT TIME of 10-15 seconds before learners move. Cold-call 4 learners to read a question and the answer they added, and ask the class: 'What evidence supports this answer?' Ask: 'Which question</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Driving Question Board revisit (Tracking understanding over time). Learners match evidence to the original questions and see how far the class has come. This builds understanding because they can see their questions turning into explanations and can identify what still needs investigation.</p> | <p>Check that sticky notes (1) contain an answer with a number or a rotation word, (2) connect to the right question, and (3) include at least one new, testable question. Tally how many questions are marked answered, partly answered and open. Use the tally to decide what to emphasise in</p>                                                                                                                                                                                                                                 |

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| <p>for this lesson.<br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/><a href="#">Determine rotations</a><br/>Related topic: not an exact match for this lesson.<br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES</a> for similar readings</p>                                                                                                                                  | <p>evidence from today (for example, 'Earth rotates 360° in 24 h, so 15° per hour; Mombasa is 4.9° further east than Kisumu, so about 20 minutes earlier'). They mark questions as 'answered', 'partly answered' or 'still open' and add at least one new question on a yellow note (for example, 'Why does the length of day change a little through the year?' or 'Do places in Turkana see sunrise at a different time from Nairobi?').</p>                                                                                                                                                                                                                                                                                                                        | <p>is still open, and what could we do in Lesson 10 to answer it?' Point out that a good DQB question may take several lessons to answer fully. Ensure that every learner places at least one sticky note and that the answers use the language of rotation.</p>                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                             | <p>Lesson 10.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p><b>Model Building Phase</b><br/> <b>VIDEO:</b><br/><a href="#">Rotating points</a><br/>Related topic: not an exact match for this lesson.<br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/><a href="#">Determine rotations (basic)</a><br/>Related topic: not an exact match for this lesson.<br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES</a> for similar readings</p> | <p>Learners return to the Lesson 1 model and revise it on a fresh page with a labelled diagram: Earth viewed from above the North Pole with an anticlockwise arrow, the axis marked as the centre, the torch (Sun) on one side, and Mombasa, Nairobi and Kisumu marked as points on the globe. They add three labels: 'centre: Earth's axis', 'angle: 15° per hour (360° in 24 h)', 'direction: anticlockwise (positive)'. They add a small table showing the longitude differences and time differences (2.9° = about 12 min, 2.0° = about 8 min, 4.9° = about 20 min). They write two sentences underneath explaining day and night, and one sentence that answers Wanjiku: 'Kenya is not still; it turns with Earth.' Finally they complete the summary table.</p> | <p>Say: 'Compare your new model with the first one. What did you add or change, and why?' Give WAIT TIME of 10-15 seconds, then ask learners to do a 3-minute gallery walk to check one other group's model against a checklist (centre, angle, direction, three cities, time differences). Cold-call 3 learners to name one improvement they noticed in a partner's model. Ask: 'Where does your model still not explain something?' (for example, the tilt and seasons) and record these as ideas for later. Close by asking: 'How would you describe the turning of Earth to a learner in Lodwar in one sentence using centre, angle and direction?'</p> | <p>Cumulative model revision with a gallery walk (Developing and Using Models). Learners compare the old and new models and evaluate peers' work against a checklist. This builds understanding because revising the model forces them to integrate the mathematics of rotation with the phenomenon and recognise gaps.</p> | <p>Check that revised models (1) include an anticlockwise direction arrow, (2) label Earth's axis as the centre, (3) show 15° per hour, and (4) show Mombasa, Nairobi and Kisumu in the correct east-to-west order with the time differences. Observe whether learners can explain a change they made. Collect the revised models and sort them into 'complete', 'missing one element' and 'needs support' to plan quick feedback at the start of Lesson 10.</p> |

## D. TEACHER REFLECTION

After the lesson, reflect on these questions: Did learners use centre, angle and direction accurately when describing Earth's turning? Which learners still said that the Sun moves around Kenya, and what evidence from the globe model helped them change their minds? Were the calculations of 12, 8 and 20 minutes secure, or did

some learners confuse longitude differences with times? Did the WAIT TIME and cold-calling bring in quieter learners? What would you change in the group roles at the globe station? Record which DQB questions remain open and what to emphasise in the Lesson 10 performance task.

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) |                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What did I observe?                                          | We observed that when the globe is turned anticlockwise 15° for each hour with the torch fixed, the Mombasa dot enters the light first, then Nairobi, then Kisumu. The time-lapse also showed dawn over Mombasa, midday over Nairobi, sunset over Lake Victoria and darkness over Kisumu.    |
| What did I learn?                                            | We learned that Earth rotates 360° in 24 hours, which is 15° per hour, anticlockwise when viewed from above the North Pole, about a centre that is Earth's axis. The longitude differences of 2.9°, 2.0° and 4.9° give time differences of about 12, 8 and 20 minutes.                       |
| How does this explain the phenomenon?                        | Day and night repeat every 24 hours because Earth rotates once about its axis in 24 hours while the Sun lights one half. Mombasa is further east than Kisumu, so it turns into the sunlight about 20 minutes sooner. The Sun only appears to move east to west; Kenya is turning with Earth. |

## LESSON 10 (80 minutes (double lesson)): Consolidation and Performance Task: Designing a Rotationally Symmetric Pattern and the Final Explanation of Day and Night

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>Purpose</b>              | Learners consolidate the whole sub-strand by (1) designing, drawing and explaining a rotationally symmetric pattern as a performance task, and (2) reviewing the DQB and their Lesson 1 model to write a Final Explanation of why Kenya has a daily pattern of day and night.                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Knowledge</b>            | By the end of the lesson the learner should be able to: state that the order of rotational symmetry is the number of times a figure fits onto itself in one full $360^\circ$ turn and that the smallest angle of fit is $360^\circ \div \text{order}$ ; describe any rotation using centre, angle and direction (positive = anticlockwise, negative = clockwise); and explain that day and night repeat every 24 hours because Earth rotates once ( $360^\circ$ ) about its axis in 24 hours, anticlockwise at $15^\circ$ per hour as seen from above the North Pole, while the Sun lights one half, so Mombasa gets sunrise about 20 minutes before Kisumu.                   |
| <b>Skills</b>               | Science and Engineering Practices: Developing and using models (design a pattern and revise the Lesson 1 globe model); Constructing explanations (write the Final Explanation using evidence); Engaging in argument from evidence (peer critique using a rubric); Using mathematics and computational thinking ( $360^\circ \div \text{order}$ , $360^\circ \div 24 \text{ h} = 15^\circ$ per hour, $4.9^\circ \div 15$ ); Obtaining, evaluating and communicating information (gallery walk). Core competencies: Creativity and Imagination, Critical Thinking and Problem Solving, Communication and Collaboration.                                                          |
| <b>Attitudes</b>            | Learners appreciate the use of rotation and rotational symmetry in day-to-day life, from Mama Achieng's kiondo to Lake Turkana wind turbines, show Responsibility in using instruments (compasses, protractors, tracing paper) with care, show Unity in giving kind and specific peer feedback, and show self-esteem (Life Skills) in presenting their explanations.                                                                                                                                                                                                                                                                                                           |
| <b>Key Inquiry Question</b> | 1. How do we use rotation in real-life situations? 2. Where do we apply rotational symmetry in day-to-day life?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Purpose in Storyline</b> | This is the final lesson of the storyline. Lesson 1 launched the phenomenon (day and night sweeping over Mombasa, Nairobi, Lake Victoria and Kisumu) and a first globe-and-torch model. Lessons 2 to 4 gave the properties of rotation, finding centre and angle, and grid rotations with positive and negative angles. Lessons 5 to 8 covered the order of rotational symmetry of plane figures, axes of solids, congruence and real-life applications. Lesson 9 revisited the phenomenon using $15^\circ$ per hour and longitudes. Lesson 10 pulls these together in a performance task (design and explain a symmetric pattern) and the Final Explanation, closing the DQB. |
| <b>Safety Notes</b>         | Compass points and pencils are sharp: carry them point down and keep them on the desk when not in use. The torch is used only for the globe model; do not shine it into anyone's eyes. Keep the walkways clear during the gallery walk. Hand out grid paper, tracing paper and protractors from the front and collect them at the end.                                                                                                                                                                                                                                                                                                                                         |

### B. LESSON OVERVIEW

Teacher Mwangi opens by replaying the 24-second time-lapse (1 second = 1 hour =  $15^\circ$  of turning) and asking learners to look at it as mathematicians: where is the centre, what is the angle and what is the direction? Learners first predict how much of their own growth they can show. They then spend most of the lesson on a performance task: individually or in pairs they design a rotationally symmetric pattern (inspired by Mama Achieng's kiondo with order 6, Mzee Kamau's 8-spoke wheel, or the 3-blade Lake Turkana turbines) on grid or plain paper, state its centre, order and smallest angle ( $360^\circ \div \text{order}$ ), and check it by tracing and turning. Learners present in a gallery walk and give feedback with a rubric. In the Explain phase they connect the pattern task to the phenomenon: Earth's turning is also described by a centre (axis), an angle ( $15^\circ$  per hour) and a direction (anticlockwise). They review the DQB, mark questions answered with evidence, and complete the Final Explanation. Finally they place their Lesson 1 model beside a revised model and write what changed, ending with the summary table. The final outcome: day and night repeat every

24 hours because Earth rotates once (360°) about its axis in 24 hours, anticlockwise at 15° per hour as seen from above the North Pole, while the Sun always lights one half. Mombasa turns into sunlight about 20 minutes before Kisumu, and the Sun only appears to move east to west.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Intro to rotational symmetry</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar videos<br><br> <b>EXERCISE:</b><br><a href="#">Rotational symmetry of 2D shapes</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar readings | Learners watch the 24-second time-lapse again (dawn Mombasa about 6:30 am, midday Nairobi about 12:30 pm, sunset Lake Victoria about 6:40 pm, darkness over Kisumu about 7:30 pm). On a sticky note or in their exercise book they individually write: (a) one prediction of what a rotationally symmetric pattern needs (centre, angle, order) and (b) a one-sentence prediction of the final answer to 'What makes it possible for a daily pattern of day and night on Earth?' They then share with a partner. Wanjiku's comment 'Kenya feels still' is put on the board and learners say whether they agree now.                          | Say: 'Today we finish our story. Watch the video one more time, but this time look for a centre, an angle and a direction.' Play the video once. Say: 'Silently write for 2 minutes: what do you now think makes day and night repeat every 24 hours, and what three things do you need to describe any turn?' Give WAIT TIME of 10-15 seconds after asking: 'Wanjiku says Kenya feels still. Do you agree with her now?' Cold-call 4 learners (not volunteers) to share predictions and record key words on the board without judging. Ask: 'What would a rotationally symmetric pattern need so that it fits onto itself?' Cold-call 2 more learners. Announce the performance task and show the rubric chart. | Predict-and-compare (Think-Pair-Share): writing a prediction before the task makes current thinking visible so that at the end learners can compare it with their final explanation. Cold-calling ensures every learner is accountable for ideas, and pairing lets them rehearse the vocabulary of centre, angle and direction.                                                                     | Observe whether predictions mention rotation or turning of Earth rather than the Sun moving; note learners who use centre, angle or direction vocabulary and those who confuse 'rotation' with 'revolution' or who say Kenya is stationary. Tick on a class list the learners who mention 360° or 24 hours. Use this to decide which pairs get an early check-in during the performance task.                                                                                                                                                                         |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Intro to rotational symmetry</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar videos<br><br> <b>EXERCISE:</b><br><a href="#">Rotational symmetry of 2D shapes</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a>              | Learners complete the performance task: Design a rotationally symmetric pattern. Step 1: choose a context (Mama Achieng's kiondo star pattern with order 6, Mzee Kamau's 8-spoke bicycle wheel with order 8, a matatu wheel with 5 nuts, order 5, or Lake Turkana turbine with 3 blades, order 3, or their own). Step 2: choose a centre of rotation and an order, and calculate the smallest angle as $360^\circ \div \text{order}$ (e.g. $360^\circ \div 6 = 60^\circ$ , $360^\circ \div 8 = 45^\circ$ , $360^\circ \div 5 = 72^\circ$ , $360^\circ \div 3 = 120^\circ$ ). Step 3: draw one motif, then rotate it about the centre through | Circulate with a checklist. Ask: 'Where is your centre? How did you decide the angle?' and 'If your order is 6, why is the smallest angle 60° and not 30°?' Wait 10-15 seconds for answers before prompting. Say: 'Show me how you know the pattern fits onto itself in one full turn of 360°.' Challenge advanced pairs: 'Can you make a pattern with order 1? What would that look like and does it have rotational symmetry?' (order 1 means no rotational symmetry). Cold-call 3 pairs mid-task to share how they checked with tracing paper. Remind them: 'Positive                                                                                                                                         | Performance task with self-check by testing (modelling in action): learners apply a rule they learned (order and angle = $360^\circ \div \text{order}$ ) to create something new, and the tracing paper test gives them evidence about whether their pattern really has rotational symmetry. Designing, rather than only identifying, forces them to reason from the idea that a full turn is 360°. | Observable indicators: the order stated matches the number of identical motifs placed evenly around the centre; the angle equals $360^\circ \div \text{order}$ with correct arithmetic; the learner states a centre and a direction; the tracing test is actually demonstrated and the count of fits is correct. Common errors to note: motifs spaced unevenly, angle calculated as $\text{order} \div 360$ , forgetting the centre, or an unsymmetrical motif that gives order 1. Record each pair's status as secure, developing or needs support on the checklist. |

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| <p><a href="#">for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>the angle repeatedly (using protractor, tracing paper or compass) until the full turn is complete. Step 4: test the pattern by tracing it and turning the tracing paper about the centre, counting how many times it fits in <math>360^\circ</math>. Step 5: write the centre, angle (state positive for anticlockwise), order and one sentence on where this symmetry is used in real life. Pairs may also compute one rotation of a point on the Cartesian plane as a bonus, such as A(2,1) rotated <math>90^\circ</math> anticlockwise about O(0,0) to A'(-1,2) using the rule (x,y) to (-y,x).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>angle means anticlockwise, negative means clockwise, so <math>-60^\circ</math> is <math>60^\circ</math> clockwise.' Keep a visible 30-minute timer and prompt every 10 minutes. Do not re-teach order or angle finding; direct learners to their Lesson 5 notes and Lesson 4 rules.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Determining rotations</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Determine rotations</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Part A (gallery walk and peer review, about 12 minutes): learners place their patterns on desks, move round in groups of 4, and use the rubric (centre, order, angle, accuracy of test, real-life link) to give each design 'one star and one wish'. Part B (about 12 minutes): the class connects patterns to the phenomenon. Using the globe and torch, learners turn the globe anticlockwise through <math>15^\circ</math> for each hour (viewed from above the North Pole) and describe Earth's rotation with centre (Earth's axis through the poles), angle (<math>360^\circ</math> in 24 h = <math>15^\circ</math> per hour) and direction (anticlockwise). They explain why Mombasa (<math>39.7^\circ</math> E) gets sunrise about 20 minutes before Kisumu (<math>34.8^\circ</math> E): <math>39.7 - 34.8 = 4.9^\circ</math>, and <math>4.9 \div 15</math> is about 0.33 h, about 20 min, with Nairobi (<math>36.8^\circ</math> E) in between. Each learner writes the Final Explanation in 5 to 7 sentences using the sentence frame: 'Day and night repeat every 24 hours because Earth ____ once (____<math>^\circ</math>) about its ____ in ____ hours,</p> | <p>For the gallery walk say: 'Walk quietly, read each rubric, and write one star (something done well) and one wish (one improvement) on a sticky note.' After 12 minutes ask: 'Which pattern had the most surprising order? What real-life object uses that order?' Cold-call 3 learners and wait 10-15 seconds before each answer. Move to the globe: 'Turn the globe through <math>15^\circ</math>. How many hours has passed? Through <math>90^\circ</math>? Through <math>360^\circ</math>?' Cold-call 4 learners to answer and demonstrate. Ask: 'What is the centre, angle and direction of Earth's turning?' Then: 'Wanjiku says Kenya feels still. Use evidence to respond.' Wait 10-15 seconds, then call on 3 learners. Ask for the time difference with: 'Calculate <math>4.9 \div 15</math> and tell me what it means for Mombasa and Kisumu.' Circulate while learners write the Final Explanation; do not give answers, ask: 'What evidence supports that sentence?' Select two explanations (one strong, one partial) to read aloud anonymously and ask the class to improve the</p> | <p>Claim-Evidence-Reasoning with a sentence frame: learners state a claim (Earth rotates once in 24 h), cite evidence (<math>15^\circ</math> per hour, longitudes, 20 minute difference, globe model, video snapshot times) and give reasoning (the Sun lights one half, so places turn into light in order from east to west). Peer 'star and wish' review builds argument-from-evidence skills and shows that the same maths ideas of centre, angle and direction transfer from patterns to Earth.</p> | <p>Check the Final Explanation against four indicators: (1) states Earth rotates <math>360^\circ</math> in 24 h at <math>15^\circ</math> per hour; (2) states anticlockwise from above the North Pole; (3) explains that the Sun lights one half and that Mombasa gets sunrise about 20 minutes before Kisumu because it is further east; (4) says the Sun only appears to move. Note misconceptions such as 'the Sun goes round Earth each day', 'Earth's revolution causes day and night' (rotation is the correct word), or 'Kenya is stationary'. Use the star and wish notes as evidence of peer feedback quality. Give a quick 3-question exit check: order of a regular hexagon (6) and its angle (<math>60^\circ</math>), degrees Earth turns in 2 hours (<math>30^\circ</math>), and direction of Earth's turn (anticlockwise).</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | turning ____ at ____° per hour. The Sun's light always lights _____. Places further east like ____ turn into sunlight ____ than places further west like ____, so _____. The Sun only ____ to move east to west.'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | partial one using the frame.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Driving Question Board (DQB) Creation</b><br> <b>VIDEO:</b><br><a href="#">Rotating points</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar videos<br><br> <b>EXERCISE:</b><br><a href="#">Determine rotations (basic)</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar readings | Learners gather at the wall chart DQB for a final review. In groups of 4, each group takes 3 or 4 questions (for example: What keeps the cycle regular? Why does the Sun's position change? Why does Mombasa get sunrise before Kisumu? How can the turning be described mathematically with a centre, an angle and a direction?). For each question they use a coloured sticky note to write the answer and the evidence (lesson and data), then place it beside the question. Questions that remain unanswered are marked and new questions are added (e.g. Why is Earth tilted about 23.5°? Why do seasons occur?). Each group reports one question that changed most for them. The class completes the sentence 'At the start I wondered _____ and now I know _____ because _____'. | Say: 'Our DQB has been growing since Lesson 1. Each group, choose the questions you can now answer and give me evidence, not just opinion.' Prompt: 'Which lesson gave you evidence for this? Was it the 15° per hour, the longitudes or the globe model?' Wait 10-15 seconds, then cold-call 1 member from each group (about 5 to 6 learners) to read an answer. Ask: 'Which question is still open? What new question has our learning raised?' Take 3 new questions and add them in a different colour for future learning. Ask a reflective question: 'How did your thinking about Kenya being still change?' Cold-call 3 learners. Keep the DQB on the wall for the following topic, and photograph it for the class record if a device is available. | Driving Question Board review (public record of growth): attaching evidence to each question helps learners see that science and mathematics knowledge is built step by step and that unanswered questions are the next start. Naming the lesson where evidence came from reinforces the storyline and shows how one idea builds on the previous one.        | Look for notes that cite specific evidence (e.g. 'Lesson 9: $4.9^\circ \div 15$ is about 20 min') rather than only copying statements. Count the number of original DQB questions answered with evidence by each group and note any left unanswered. Observe whether learners ask new, deeper questions (such as tilt or seasons), which shows curiosity. Listen for accurate vocabulary in the sentence 'At the start I wondered... now I know...'.                            |
| <b>Model Building Phase</b><br> <b>VIDEO:</b><br><a href="#">Examples: Determine Angles of Rotation</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: CK-12<br> <a href="#">Search ARES</a> for similar videos                                                                                                                                                                                                                                                                                                                                                                                                                                      | Learners place their Lesson 1 initial model (drawing or notes of the globe and torch) beside a new final model. The final model is a labelled diagram showing: Earth's axis through the North and South Poles as the centre of rotation, a curved arrow showing anticlockwise rotation viewed from above the North Pole, the 15° per hour rate and 360° in 24 h, the Sun's light lighting one half, and Mombasa, Nairobi and Kisumu marked in east-to-west order with                                                                                                                                                                                                                                                                                                                   | Say: 'Take out your Lesson 1 model. What is missing from it?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners. Say: 'Now draw your final model. Include a centre, an angle and a direction, and show where Mombasa, Nairobi and Kisumu are.' Circulate and ask: 'How does your model show 15° per hour? Where is Mombasa in sunlight before Kisumu?' Ask a pair to use the globe and torch to demonstrate the final model while the class                                                                                                                                                                                                                                                                                                       | Model comparison (Then and Now) and revision: placing the first and final models side by side makes learning visible and lets learners judge how well their model explains the phenomenon. Adding the mathematical elements of centre, angle and direction shows that the model has moved from a drawing to a quantitative explanation that predicts timing. | Use a 4-point checklist for the final model: axis labelled as centre; anticlockwise direction shown; 15° per hour or 360° in 24 h stated; east-to-west order of Mombasa, Nairobi, Kisumu with the Sun lighting one half. Check that the Then and Now comparison names at least one specific improvement (e.g. 'the first model did not explain why Mombasa gets sunrise first'). Review the summary table for accuracy and completeness, and note learners needing follow-up on |

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| <p> <b>HTML:</b><br/> <a href="#">Angles of Rotation in Standard Positions: Clock Angles (PLIX)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>times (about 20 minutes between Mombasa and Kisumu sunrise). They add a small inset of their symmetric pattern, showing that both Earth's turning and the pattern are described with a centre, an angle and a direction. They then write 3 sentences in a 'Then and Now' comparison: what the Lesson 1 model could not explain, what the final model explains, and what they would still like to investigate. Finally, they fill in the summary table (What we observed, What we learned, How we can explain it).</p> | <p>checks it against the rubric. Ask: 'Which part of your model is the most important change?' Cold-call 4 learners. Close with: 'Mama Achieng turns a motif 60° six times to finish a kiondo. How is Earth's turn like that, and how is it different?' (Earth turns continuously at 15° per hour; the pattern is turned in steps.) Collect models and exit tickets, and thank learners for their effort.</p> | <p>the difference between rotation and the Sun's apparent motion.</p> |
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## D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners use centre, angle and direction fluently, and could each state the order and the angle ( $360^\circ \div \text{order}$ ) for their pattern correctly? (2) How many Final Explanations contained all four indicators, and which misconceptions persisted (Sun moves round Earth, rotation/revolution confusion, Kenya is still)? (3) Did the performance task allow every learner to show success, including those who struggled with calculation or drawing, and what support (tracing paper, pre-drawn templates) worked well? (4) Which DQB questions remain open and how will they feed into future lessons? (5) Which evidence from the lesson (checklists, star and wish notes, exit tickets, models) will be used to give feedback and plan remediation or extension, for example order 1 versus no symmetry, or the cube as an extension?

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What did I observe?</b>                   | What we observed: In the time-lapse, dawn came to Mombasa (about 6:30 am) first, then midday over Nairobi (about 12:30 pm), sunset over Lake Victoria (about 6:40 pm) and darkness over Kisumu (about 7:30 pm), and the pattern repeated every 24 hours. We also observed that our patterns fit onto themselves several times in one full turn, and that the globe with a torch lit one half at all times as it turned.                                                                                     |
| <b>What did I learn?</b>                     | What we learned: A rotation is described by a centre, an angle and a direction (positive = anticlockwise, negative = clockwise). The order of rotational symmetry is the number of times a figure fits onto itself in $360^\circ$ , and the smallest angle is $360^\circ \div \text{order}$ (e.g. kiondo order 6 at $60^\circ$ , 8-spoke wheel order 8 at $45^\circ$ , Lake Turkana turbine order 3 at $120^\circ$ ). Earth turns $360^\circ$ in 24 hours, which is $15^\circ$ per hour.                    |
| <b>How does this explain the phenomenon?</b> | How we can explain it: Day and night repeat every 24 hours because Earth rotates once ( $360^\circ$ ) about its axis in 24 hours, turning anticlockwise at $15^\circ$ per hour as seen from above the North Pole, while the Sun always lights one half. Mombasa is further east than Kisumu ( $4.9^\circ$ difference, $4.9 \div 15$ is about 20 minutes), so it turns into sunlight about 20 minutes sooner. The Sun only appears to move east to west, and Kenya is not still because it turns with Earth. |



| DIFFERENTIATION AND INCLUSION       |                                                                                       |                                                                                  |
|-------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Learner Need                        | Support Strategies                                                                    | Extension Strategies                                                             |
| English Language Learners           | Provide bilingual glossaries; allow use of first language for initial model drawing.  | Write extended explanations of observations; lead group discussions in English.  |
| Students with learning difficulties | Provide structured note frames; pair with a supportive partner during investigations. | Mentor peers; create additional visual models to explain concepts.               |
| Gifted and advanced learners        | Access to additional reading materials; challenge questions during DQB.               | Lead model-building discussions; research and present extension topics to class. |